

Non-Thermal Electromagnetic Fields as a Selective Antitumor Monotherapy

A cross-study synthesis of 49 in vitro investigations on field type, dose, and selectivity

Senior Biomedical Researcher Report

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Literature-based meta-analysis · 49 papers · 434 non-thermal experimental conditions

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Abstract

Background. Non-thermal electromagnetic fields (EMFs) have been proposed as a selective antitumor modality across a broad range of physical configurations, from extremely-low-frequency (ELF) sinusoidal magnetic fields in the μT –mT range, to intermediate-frequency alternating electric fields (TTFields, 100–300 kHz), to ultrashort high-amplitude pulsed electric fields (nsPEF), and to millimetre-wave (MMW) and terahertz radiation. The mechanistic plausibility of a non-thermal cancer-selective effect rests on the differential biophysical properties of malignant cells (membrane polarisation, dielectric anisotropy, mitotic spindle dynamics, redox imbalance), but the literature is fragmented and exposure parameters are highly heterogeneous.

Methods. The provided dataset of 439 experimental conditions extracted from 49 in vitro studies (Q1 = 307 conditions; Q2 = 127) was filtered to retain only non-thermal exposures (Thermal_Mechanism = 0), yielding 434 conditions across all 49 papers. Field_Type labels were rationalised into 11 physics-based macro-categories (A–K). Because macro K (ELF-EMF general) was the largest and most heterogeneous group, it was further partitioned into three waveform-and-amplitude sub-modules — K1 (sinusoidal at physiological amplitudes), K2 (pulsed/square at higher amplitudes), and K3 (specialty / under-specified waveforms) — yielding a final 13-module analytical structure. Within each module, exposure parameters, cell-fate outcomes (viability, proliferation, apoptosis, necrosis, senescence, motility, ferroptosis), mechanistic outcomes (ROS, antioxidant response, mitochondrial membrane potential, calcium, morphology/cytoskeleton, metabolic reprogramming, cell cycle), and molecular outcomes (genomic, proteomic) were analysed both in cancer (Is_Cancer=1) and matched healthy (Is_Cancer=0) cells, with explicit reporting of the number of independent papers underlying each statistic.

Findings. Three macro-categories converge on the strongest combination of antitumor efficacy and selectivity: (i) intermediate-frequency alternating magnetic fields (AMF, 100–320 kHz, 1.4–18 mT) — Paper 4 reduces viability of 9 cancer cell lines (48–88 % vs control) at 227 kHz / 18 mT while leaving 3 healthy lines unaffected (NHA, HCF, HUVEC: 90–105 %); (ii) TTFields (150 kHz, 1.5 V/cm) — strong proliferation inhibition in cancer (Direction = -1 in 17/19 measured cancer conditions) with partial sparing of MCF-12 below the dielectrophoretic threshold of ~ 3 V/cm; (iii) MMW exposure in the 75–105 GHz W-band at 0.2 mW/cm^2 — apoptosis induction in H1299 lung cancer cells (Apoptosis %vs control up to 1542 %) coupled with senescence in 6/7 cancer conditions, while 15/15 MCF-10A conditions remained at Direction = 0 for all measured outcomes. Conversely, ELF-EMF exposures around 50 Hz showed inconsistent selectivity (Paper 19: hormetic biphasic response in MCF-10A; Paper 1: cytotoxicity also in healthy C8D1A astrocytes), and LTE-modulated UHF radiation reduced proliferation indistinguishably in cancer and IMR-90 fibroblasts.

Conclusions. The evidence supports a working hypothesis that the most therapeutically attractive non-thermal window for monotherapy lies in the intermediate-frequency band (~ 100 kHz to ~ 300 kHz), regardless of whether the dominant component is electric (TTFields) or magnetic (AMF), because cancer-

cell sensitivity in this band appears to derive from disruption of mitotic-spindle alignment and mitochondrial membrane potential, two features that in non-dividing/post-mitotic healthy cells are minimally engaged. ELF (≤ 100 Hz) regimes are more compatible with adjuvant or sensitiser roles than with monotherapy. A future experimental hypothesis is articulated: 200–230 kHz sinusoidal magnetic field at 5–18 mT applied for 30–60 min, tested on a panel of glioblastoma and triple-negative breast cancer lines paired with isogenic non-tumorigenic controls, with primary readout on mitochondrial membrane potential ($\Delta\Psi_m$) and γ -H2AX, and selectivity assessed against NHA, MCF-10A and HUVEC.

Document outline

This report follows the four-step analytical structure requested: (1) preliminary filtering, macro-category construction, and gap identification; (2) thirteen self-contained subgroup modules covering the 11 macro-categories with macro K partitioned into K1/K2/K3; (3) cross-subgroup synthesis with mechanistic and selectivity comparison and an experimental hypothesis; (4) global limitations and conclusions. Eight figures (selectivity forest, frequency-band heatmap, mechanism coverage matrix, paper-level selectivity scatter, frequency $\times$ amplitude biophysical-window map, outcome-direction distribution per sub-module, macro $\rightarrow$ pathway bipartite map, cell-line-resolved selectivity) and two tables are embedded inline.

- Section 1 — Dataset structure and analytical strategy
- Section 2 — Step 1: thermal filtering, macro-category construction, gap identification
- Section 3 — Step 2: subgroup-level modules (A, B, C, D, E, F, G, H, I, J, K1, K2, K3)
- Section 4 — Step 3: cross-subgroup synthesis and experimental hypothesis
- Section 5 — Step 4: global limitations and conclusions

1. Dataset structure and analytical strategy

The provided dataset contains 439 rows $\times$ 132 columns, where each row represents a unique experimental condition (cell line $\times$ field configuration $\times$ exposure timing). The 49 contributing papers were published between 2020 and 2026 and were sourced predominantly from Q1 journals (307 of 439 rows, 70 %) with the remainder from Q2 journals (127 rows, 29 %); 5 rows had Q1 mark but came from a single bordering source. All studies are in vitro.

Columns are organised into six logical blocks: (1) metadata (Paper_ID, Year, Journal, Ranking, Cell_Line, Culture_Type, Is_Cancer, Type, Origin, Study_Type); (2) exposure parameters (Field_Type, Wave_Type, AC frequency and intensity, DC component, electric-field intensity, power density, SAR, pulse duration / repetition rate / number, Field_Notes, Device, Thermal_Mechanism, Control_Group); (3) cell-fate outcomes for six endpoints (viability, proliferation, apoptosis, necrosis, senescence, motility), each with assay, exposure duration, observation time, %_vs_Control, Direction (-1, 0, +1), and free-text notes, plus a Ferroptosis_Flag/Notes pair; (4) mechanistic outcomes (ROS, antioxidant, calcium flag, cellular morphology, mitochondrial membrane potential, metabolic reprogramming, differentiation, cytoskeleton, cell cycle); (5) molecular outcomes (Genomic_*, Proteomic_*) with assay, target family, gene/protein list, direction, mechanistic detail, and notes.

The success criteria provided in the prompt were applied as follows. For cancer cells, success means Direction = -1 for viability, proliferation, and motility; Direction = +1 for apoptosis, necrosis, senescence; ROS = +1 and Antioxidant = -1 are also success-coherent. For healthy cells, success means Direction $\in \{0, +1\}$ for viability and proliferation; Direction $\in \{0, -1\}$ for apoptosis, necrosis, senescence, and motility damage; ROS $\in \{0, -1\}$ and Antioxidant $\in \{0, +1\}$. For the remaining mechanistic blocks (membrane potential, calcium, cytoskeleton, metabolic reprogramming, ferroptosis, morphology, cell cycle, genomic, proteomic), no fixed success criterion is applied; instead, the report discusses whether an effect is present and how it connects to the cell-fate readout.

Throughout the analysis, two distinct quantities are tracked separately: number of conditions (rows) measuring an outcome, and number of independent papers behind those rows. A coverage figure dominated by a single Paper_ID is flagged as such, because a thirty-condition coverage from one prolific paper is mechanistically less robust than a ten-condition coverage from five independent papers.

2. Step 1 — Preliminary filtering, macro-category construction, and gap identification

2.1 Thermal filtering

The Thermal_Mechanism column was inspected first. Only 5 of 439 rows (1.1 %) reported a thermal mechanism (Thermal_Mechanism = 1) and were excluded. The remaining 434 rows from all 49 papers form the working dataset for the rest of the report. No paper was lost entirely by this filter; it removed only a subset of conditions from a small number of studies. From this point onward, the term “dataset” refers to these 434 non-thermal conditions.

2.2 Macro-category construction

The Field_Type column contained 27 distinct labels (after whitespace normalisation), with substantial overlap between the underlying physics (e.g. five different labels all denoting ELF-EMF in the 1–100 Hz band, three different labels denoting TTFields-class intermediate-frequency electric fields, two labels denoting telecom-band UHF radiation). Taking labels at face value would have fragmented the dataset into clinically meaningless subgroups. I therefore reorganised the 27 labels into 11 physics-based macro-categories, ordered roughly by frequency and pulse character. The mapping is shown in Table 1.

Table 1 lists the 11 macro-categories, the original Field_Type labels merged into each, the underlying physics, and the number of conditions and independent papers contributing. The 11 categories together account for all 434 non-thermal rows; no row was orphaned.

Macro-category (code)	Original Field_Type labels merged	Defining physics	n rows	n papers
A — nsPEF (ultrashort pulsed electric)	nsPEF	Sub-microsecond pulsed electric field, ns scale, 8–100 kV/cm, 1–10 000 pulses; intracellular electroporation regime	37	5
B — nsPMF / μ sPMF (ultrashort pulsed magnetic)	nsPMF; μ sPMF	ns to μ s pulsed magnetic field, mT range; induced eddy-current electric field with high dB/dt	24	1
C — TTFields class (intermediate-frequency AEF)	TTFields (AEF); Tumor Treating Induced Fields (TTIF); AEF	Alternating electric field 100–300 kHz, ≤ 6 V/cm; established mitotic-spindle disruption mechanism	52	6
D — MMW / THz (EHF radiation)	MMW (EHF-EMF); THz radiation (EHF-EMF)	Far-field radiation in the W-band (75–110 GHz) and at 140 GHz; CW or W-band sweeping; 0.2–7.4 mW/cm ²	40	4
E — LTE / UHF telecom	LTE (UHF-EMF)	1.7–1.76 GHz, WCDMA/LTE-modulated; SAR-dose-controlled, regulatory-relevant exposure	29	2

Macro-category (code)	Original Field_Type labels merged	Defining physics	n rows	n papers
F — AMF intermediate-frequency magnetic (LF-EMF)	AMF (LF-EMF); EMF (LF-EMF)	Sinusoidal magnetic field 87 kHz – 320 kHz, 1.4–18 mT; non-hyperthermic regime explored as direct cytotoxicity	38	3
G — RF MHz fields	RF-MF + SMF (MF/HF-EMF + SMF); RF-EMF (MF-EMF); REAC (RF-EMF)	Radiofrequency MHz range (0.9–7.2 MHz, 13.56 MHz), often combined with a static DC bias; nT-μT amplitudes	30	4
H — Static magnetic field (SMF)	SMF	Static (DC) magnetic field, mT range; no time variation	4	2
I — ELF-EMF combined with SMF	ELF-EMF + SMF	Sinusoidal ELF (Hz range, μT–mT) superimposed on a DC field, often resonance-tuned (e.g. 14.5–16.5 Hz, Larmor of Ca ²⁺ /K ⁺)	12	3
J — Rotating / spinning / gradient ELF magnetic	Spinning Oscillating MF; Gradient RMF; LF-RMF; Oscillating MF (Helmholtz, fixed axis)	Rotating or strongly inhomogeneous low-Hz magnetic fields (2.5–272 Hz, 0.04–0.41 T); spatial-gradient mechanism	20	3
K — ELF-EMF general (sinusoidal & pulsed, single-axis)	ELF-EMF; ELF-MF; MEMP; PEMF; ELF-PEMF; SREMF; ELF P-EMF	Single-axis sinusoidal or pulsed ELF magnetic field, 0.01 Hz–80 Hz, μT–T amplitudes; the largest and most heterogeneous group	148	18

Table 1. Mapping of original Field_Type labels onto the 11 physics-based macro-categories used throughout the report. Note the strong imbalance in evidence base: macro K alone accounts for 148 / 434 conditions (34 %) and 18 / 49 papers (37 %), while H accounts for only 4 conditions and 2 papers.

2.3 Gap identification

Before entering the subgroup analyses, Table 2 cross-tabulates each macro-category against the eight cell-fate and direct-mechanistic outcomes for which a Direction column is available. The cell value is the number of conditions in which the Direction was reported as -1, 0, or +1 (i.e. excluding “Not performed” and “Mentioned”). The right-most column gives the number of independent contributing papers, which is the figure that constrains the strength of any conclusion.

Macro	Viab	Prol	Apop	Necr	Senesc	Motil	ROS	AOx	Papers
A — nsPEF	15	5	5	0	0	2	2	0	5
B — nsPMF / μsPMF	24	0	0	0	0	0	0	0	1
C — TTFields	32	19	5	0	0	8	1	0	6
D — MMW / THz	12	0	26	4	22	0	0	0	4
E — LTE	13	16	7	0	0	4	0	0	2
F — AMF kHz	38	2	8	8	0	0	3	9	3
G — RF MHz	10	3	6	0	0	3	16	0	4
H — SMF	2	2	0	0	0	0	0	1	2

Macro	Viab	Prol	Apop	Necr	Senesc	Motil	ROS	AOx	Papers
I — ELF + SMF	3	11	2	0	0	2	5	2	3
J — Rotating ELF	5	1	2	0	0	8	4	0	3
K — ELF general	107	100	40	18	3	7	24	1	18

Table 2. Outcome × macro-category coverage matrix. Cell values are the number of conditions in which a directional readout was reported; the rightmost column gives the number of independent papers behind the row. AOx = Antioxidant.

Five gap patterns emerge from Table 2. First, the macro-categories B (nsPMF / μ sPMF), H (SMF) and J (rotating ELF) rest on a very thin paper base (1, 2 and 3 papers respectively); any apparent uniformity of effect within them is uniformity of one to three research groups, not of the field. Second, ferroptosis and senescence are systematically under-investigated (senescence has substantive coverage only in macro D — MMW/THz, where it is the dominant readout; ferroptosis is essentially never measured outside macro I, with only 2 conditions in 1 paper). Third, antioxidant response is virtually absent in 9 of 11 macro-categories; the only macro-category in which a meaningful antioxidant comparison can be made is F (9 conditions, paper 4). Fourth, motility — the cell-fate readout most relevant to metastatic risk — is well covered only in C (TTFIELDS, 8 conditions across 4 papers) and J (rotating ELF, 8 conditions, 1 paper); for ELF-general the data are extremely sparse (7 conditions across the 18 papers in K). Fifth, the cancer-to-healthy ratio is highly skewed: of the 49 papers, only 26 (53 %) include at least one healthy cell line, and several macro-categories (B, G, H) contain zero healthy-cell conditions, which makes selectivity assessment within those categories impossible from the dataset itself.

These gaps have direct consequences for the strength of conclusions that can be drawn in each subgroup module. Where coverage is thin or single-paper-driven, the module flags this in its limitations sub-section and refrains from generalising.

3. Step 2 — Subgroup-level analysis modules

3.1 Module A — nsPEF (ultrashort pulsed electric fields)

A.1 Subgroup overview

Module A comprises 37 experimental conditions from 5 independent papers (Paper_IDs 42, 43, 45, 46, 47), of which 34 are on cancer cells and 3 are on healthy cells (all MCF-10A breast epithelial cells, from Paper 45). The cancer cell lines tested are MDA-MB-231 (triple-negative breast adenocarcinoma, human), Hep3B (hepatocellular carcinoma, human), MCF-7 (luminal breast adenocarcinoma, human), U-87 MG (glioblastoma, human), C6 (rat glioma), GL261 (mouse glioma), A375 (melanoma, human), C32 (melanoma, human), and others. The subgroup is dominated by glioblastoma and breast cancer; melanoma is represented but in fewer conditions. Only Paper 45 enables a direct selectivity assessment, on a single isogenic-style pair (MCF-7 vs MCF-10A). The cancer-to-healthy ratio (34 : 3) is too unbalanced for any robust population-level selectivity claim within this macro-category.

A.2 Exposure parameters

The waveform is a square pulse with sub-microsecond duration in all five papers. Pulse durations span 5 ns, 50 ns, 100 ns and 300 ns. Electric-field amplitudes span almost two orders of magnitude: 8, 10, 15, 20, 40, 70, 80, 90 and 100 kV/cm. Pulse repetition rates range from 2 Hz (slow) to 200 kHz (very fast); pulse counts from 60 to 10 000. Devices are typically lab-built nanosecond pulse generators. The most distinctive parameter is the high field amplitude, which places these exposures in the intracellular electroporation regime (membrane potential rise on the timescale of pulse rise-time, i.e. ns).

Frequency rationales were not relevant for nsPEF (no carrier frequency); intensity rationales were uniformly empirical (titrating amplitude × pulse number until a viability effect was reached). The spread of parameters tested in this dataset is therefore broad on amplitude but narrow on pulse-shape diversity.

A.3 Cell-fate outcomes

Viability was measured in 15 conditions (cancer = 12, healthy = 3). Of the 12 cancer-cell viability measurements, 6 reported Direction = -1 (50 %), 6 reported Direction = 0 (50 %), and none reported a pro-proliferative effect. The healthy-cell viability measurements (Paper 45, MCF-10A) all gave Direction = 0 across 10–20 kV/cm at 50 ns pulse duration; Paper 45 simultaneously reported Direction = -1 for MCF-7 at 15 and 20 kV/cm under the same pulse parameters. This is the cleanest selectivity demonstration in this macro-category, but it is restricted to a single paper, a single pulse duration (50 ns), and a single isogenic-style pair.

Proliferation was measured in 5 conditions (cancer only): 4 reported Direction = -1 and 1 reported Direction = 0. Apoptosis was measured in 5 conditions (cancer only): 4 reported Direction = +1 and 1 reported Direction = 0; the +1 dominance is consistent with the working assumption that intracellular

electroporation triggers programmed cell death rather than mere lysis. Necrosis was not directly measured. Motility was measured in 2 conditions (cancer only): one Direction = -1, one Direction = 0; the dataset is too small to draw a conclusion. Senescence and ferroptosis were not measured in this macro-category.

A.4 Mechanistic outcomes

The most striking mechanistic signature in this subgroup is the mitochondrial membrane potential collapse. Direction = -1 was reported in 21 of the 21 conditions in which it was measured — i.e. every nsPEF condition that probed $\Delta\Psi_m$ reported a significant decrease, with no exceptions. This mechanistic constancy strongly supports the model in which sub-microsecond pulses create transient pores in inner-membrane lipid bilayers preferentially because of the high cumulative dB/dt and the resulting transmembrane potential transient. ROS was measured in only 2 conditions (cancer): both Direction = +1, consistent with mitochondrial leakage. Calcium involvement was flagged as +1 in 1 condition. Cellular morphology changes were observed in 29 conditions (visible blebbing, swelling, condensation), and cytoskeleton alterations in 25 conditions, supporting electroporation-driven cytoskeletal reorganisation. Cell cycle was only “Mentioned” in 3 conditions and never directly measured by phase-resolved flow cytometry within the subgroup.

A.5 Molecular outcomes

The molecular profile of macro-category A is dominated by two protein/gene families. First, MAGE (Melanoma Antigen Gene) family transcripts and the corresponding MAGE proteins, together with the immune checkpoint PD-1, were the most frequent targets — investigated in 5 conditions (Paper 47, melanoma). The reported direction is Down for MAGE family transcripts, consistent with a loss of immunoevasive antigen expression; PD-1 also down. Second, executioner caspases (Caspase-3, Caspase-7, sometimes Caspase-9) were profiled in 2 conditions, with Direction = +1, supporting apoptosis activation. Adhesion and junction proteins were also investigated in 3 conditions, with mixed direction. The molecular evidence is therefore consistent with the cell-fate observations (apoptosis + $\Delta\Psi_m$ collapse) and adds an interesting immunological dimension specific to nsPEF in melanoma.

A.6 Mechanism of action emerging within macro-category A

Synthesising the four blocks above, the recurrent mechanistic chain in the strongest cancer-killing nsPEF conditions is: pulse-induced transient transmembrane potential rise → inner-mitochondrial-membrane permeabilisation ($\Delta\Psi_m$ collapse, 100 % Direction = -1 across 21 conditions) → cytochrome-c-style apoptotic cascade (Caspase-3/7 induction in the molecular block) → caspase-driven cytoskeletal/morphological breakdown (29 morphology conditions and 25 cytoskeleton conditions reporting changes). This is mechanistically distinct from a thermal lysis pathway. The melanoma-specific MAGE/PD-1 down-regulation suggests an additional immunogenic-cell-death dimension that warrants follow-up but is currently restricted to a single paper.

A.7 Limitations of the macro-category A evidence

Three internal limitations constrain the strength of conclusions. First, the mitochondrial membrane potential coverage (21/21 Direction = -1) is dominated by Paper 47 (melanoma) and would benefit from independent replication on glioblastoma or breast cancer lines within the same macro-category. Second, the only selectivity demonstration is on a single MCF-10A paired with MCF-7 in Paper 45, with no measurement of $\Delta\Psi_m$ or apoptosis in the healthy partner; whether MCF-10A simply tolerates the field or undergoes the same mitochondrial perturbation without progressing to apoptosis is unknown. Third, the broad amplitude range (8 to 100 kV/cm) is not sampled densely enough to resolve a precise selectivity threshold, and the relationship between amplitude $\times$ pulse number $\times$ pulse duration is treated heterogeneously across the 5 papers, preventing a clean dose–response comparison.

3.2 Module B — nsPMF / μ sPMF (ultrashort pulsed magnetic fields)

B.1 Subgroup overview

Macro-category B contains 24 experimental conditions, all from a single paper (Paper 22). All 24 conditions use the same cancer cell line (rat glioma C6) and zero healthy cell line. The category is therefore mono-paper, mono-cell-line: any pattern observed within it reflects the methodology of one research group and one tumour model. No selectivity assessment is possible from the dataset itself.

B.2 Exposure parameters

Two pulse-duration regimes are explored within Paper 22. The nsPMF regime uses 600 ns square magnetic pulses at amplitudes between 0.24 and 0.787 mT, corresponding to a high dB/dt and an induced eddy-current electric field that delivers transient sub-microsecond electrical perturbation analogous to nsPEF but without electrode contact. The μ sPMF regime uses 100 μ s and 205 μ s exponential or square pulses at amplitudes between 16.4 and 854 mT — orders of magnitude higher than the ns regime, but with proportionally longer duration. Pulse repetition rate and number are not consistently reported.

The intensity rationale is empirical (titrating dB/dt against the C6 viability response). Frequency rationale is not applicable because the exposures are individual pulses rather than a continuous waveform.

B.3 Cell-fate outcomes

All 24 conditions measure viability and only viability (no proliferation, apoptosis, necrosis, senescence, motility, ferroptosis, ROS, antioxidant, cell-cycle, calcium, or molecular block was directly measured in this macro-category). Of the 24 viability measurements, 16 (67 %) reported Direction = -1 and 8 (33 %) reported Direction = 0; no condition reported a pro-proliferative effect.

The strongest reductions came from the μ sPMF regime: 67.47 % of control at 122 mT $\times$ 100 μ s and 70.8 % at 363 mT $\times$ 100 μ s. In the nsPMF regime the effects were milder (84–95 % of control). Within the nsPMF regime, the dose-response across 0.24 $\rightarrow$ 0.787 mT was non-monotonic (e.g. 84.97 $\rightarrow$ 85.9 $\rightarrow$ 85.29 $\rightarrow$ 91.86 % at 24-h observation), suggesting either a stochastic-resonance-type window or insufficient replication to resolve the curve.

B.4 Mechanistic outcomes

Within macro-category B, the cellular morphology flag is set in all 24 conditions (qualitative observation that C6 cells changed shape under exposure), but no other mechanism was probed. There are no ROS, $\Delta\Psi_m$, calcium, cytoskeleton, metabolic or molecular data in this macro-category. The mechanistic narrative is therefore restricted to qualitative morphological evidence.

B.5 Molecular outcomes

No genomic or proteomic data are reported within macro-category B.

B.6 Mechanism of action emerging within macro-category B

Given that only viability and morphology are documented, the proposed mechanism is necessarily speculative. By analogy with nsPEF (macro A), the most plausible mechanism is induced eddy-current electrical perturbation of the cell membrane and/or organelles by the very steep dB/dt of sub-microsecond magnetic pulses, leading to permeabilisation and downstream death. However, this analogy is not supported within macro-category B by any direct $\Delta\Psi_m$, ROS, or apoptotic measurement.

B.7 Limitations of the macro-category B evidence

The fundamental limitation is that this macro-category rests on one paper, one cell line, and one outcome (viability). The 67 % efficacy rate is a per-condition figure; at the paper level the figure is 1/1, which is meaningless as a base of inference. No selectivity statement is possible. No mechanistic chain can be defended from this dataset alone. The macro-category should be regarded as a proof of principle that ultrashort magnetic pulses can reduce C6 viability — and as motivating, but not establishing, a wider research programme.

3.3 Module C — TFields class (intermediate-frequency alternating electric fields)

C.1 Subgroup overview

Macro-category C comprises 52 conditions from 6 papers (Paper_IDs 17, 26, 27, 34, 39, 49) and includes 41 cancer-cell conditions and 11 healthy-cell conditions. The cancer panel covers cholangiocarcinoma (HCCC-9810, RBE), triple-negative breast cancer (MDA-MB-231, HCC38), glioblastoma (U87, U251), and melanoma (B16F10), with MDA-MB-231 the dominant model (used across multiple papers, enabling cross-paper replication). The healthy partner is MCF-12 (non-tumorigenic breast epithelial line), tested in Papers 26 and 34. The 41 : 11 cancer-to-healthy ratio is the most favourable in the entire dataset for direct selectivity assessment, and the inclusion of MCF-12 in two independent papers is one of the strongest replicability signals available.

C.2 Exposure parameters

Frequency is the defining variable: 35 / 52 conditions (67 %) used 150 kHz, 12 used 200 kHz, 3 used 100 kHz, 1 used 300 kHz, 1 not reported. The 150 kHz choice originates from the FDA-approved Optune device for glioblastoma and is repeated as a literature-derived rationale. The electric-field amplitude is the second key variable, ranging from 0.26 V/cm (well below the published therapeutic threshold) up to 6 V/cm; the typical and most-tested intensity is 1.5 V/cm. Waveform is sinusoidal in all but one condition (which used amplitude-modulated sine). Exposure is continuous over 24–72 h in most experiments. The intensity rationale is uniformly literature-derived (clinical TFields setting). The device is either the inovitro™ system (NovoCure, the commercial platform) or laboratory-built parallel-plate capacitive systems.

C.3 Cell-fate outcomes

Viability was measured in 32 conditions (cancer = 21, healthy = 11). Of the 21 cancer measurements, 12 (57 %) reported Direction = -1, 7 reported Direction = 0, and 2 reported Direction = +1 (a small pro-proliferative signal at sub-threshold amplitudes, see below). The strongest cancer-cell viability reduction was MDA-MB-231 at 150 kHz / 1.5 V/cm / 72 h continuous exposure, where Paper 26 reported Viability = 0 % of control (Direction = -1) — i.e. complete loss of viable cells. At 24 h the same condition reported 29 % of control, and at 48 h either 51 % or 74 % depending on replicate, which constitutes a clean monotonic time-dependent reduction. Paper 34 independently replicated the 150 kHz / 1.5 V/cm condition on MDA-MB-231 (24 h: 30 % of control) and HCC38 (Direction = -1).

Of the 11 healthy MCF-12 measurements, 4 reported Direction = -1, 7 reported Direction = 0; the -1 measurements were concentrated at the highest amplitude tested (6 V/cm in Paper 34). Below 4.4 V/cm, MCF-12 viability remained at 88–102 % of control, while MDA-MB-231 was already at 29 % of control. There is therefore a selectivity window: at 150 kHz, between approximately 1.5 and 4 V/cm, cancer cells

(MDA-MB-231, HCC38) show major viability reduction while MCF-12 is largely spared; above 6 V/cm the selectivity is lost.

Proliferation was measured in 19 conditions (all cancer): 17/19 (89 %) reported Direction = -1, 2 reported Direction = 0, 0 reported +1. This is the strongest and most consistent cell-fate signal in this macro-category, and it is supported by 4 of the 6 contributing papers. Apoptosis was measured in 5 conditions: 3 reported +1 (cancer), 2 reported 0; necrosis and senescence were not directly measured. Motility was measured in 8 conditions (cancer): 5 reported Direction = -1 and 3 reported 0, with the -1 results coming from MDA-MB-231 and U87.

C.4 Mechanistic outcomes

The signature mechanism of TTFields is mitotic-spindle disruption, but in the present dataset the direct measurement of cell cycle arrest is qualitative only: 20 of 41 cancer-cell conditions report Cell_Cycle_Phase = 'Mentioned' (i.e. discussed but not phase-resolved). Direct measurement of mitochondrial membrane potential is reported in only 1 condition (Direction = -1); ROS in only 1 condition (Direction = +1). Cellular morphology changes were reported in 19 conditions and cytoskeleton in 11 — both consistent with mitotic-arrest-driven changes (multinucleation, tubulin reorganisation). Calcium involvement was Mentioned in 1 condition. The mechanistic depth in this macro-category is therefore narrative-heavy and quantitatively shallow: the field is well-established by clinical experience but the molecular and biochemical details are under-measured in the included in-vitro literature.

C.5 Molecular outcomes

Proteomic data centre on damage-associated molecular patterns (DAMPs) and immunogenic-cell-death biomarkers (HMGB1, calreticulin, ATP release; 2 conditions, Paper 27 in melanoma B16F10), with Direction = +1, consistent with TTFields-induced immunogenic cell death. Genomic data are sparse within this macro-category; tubulin and septin transcript levels are mentioned but not quantitatively profiled in the dataset. The molecular evidence is more suggestive than conclusive but is consistent with the cell-fate readout.

C.6 Mechanism of action emerging within macro-category C

The synthesised mechanism within macro-category C is: 100–300 kHz alternating electric field, at amplitudes around 1.5 V/cm, causes electric-dipole alignment forces on tubulin and septin during mitosis, producing spindle misassembly → multinucleation → mitotic catastrophe → apoptosis (caspase-3 cleavage products, +1 in apoptotic readouts) → DAMP release → potential immunogenic cell death (HMGB1, calreticulin, ATP). The selectivity arises mechanistically because non-dividing healthy cells (MCF-12 in vitro reaches confluence at 4–5 days) have a much smaller fraction of cells in mitosis at any given time, so the TTFields hit is statistically weaker on a per-population basis. The empirical selectivity window observed in the dataset (1.5–4 V/cm at 150 kHz) is fully consistent with this mechanism.

C.7 Limitations of the macro-category C evidence

The principal limitation is the narrow focus on 150 kHz and the under-sampling of the alternative carrier frequencies (200 and 100 kHz). The mechanistic depth is also asymmetric: viability and proliferation are richly sampled, but $\Delta\Psi_m$, ROS, calcium and cell-cycle phase resolution are weakly measured. The healthy-line panel is a single line (MCF-12), which is informative but does not allow generalisation to non-breast tissues. Finally, several Paper 34 conditions report Viability_%_vs_Control = 'Not reported' even when Direction is reported, which limits the ability to perform fine-grained dose–response analysis.

3.4 Module D — MMW / THz (extremely-high-frequency radiation)

D.1 Subgroup overview

Macro-category D comprises 40 conditions from 4 papers (Paper_IDs 11, 12, 32, 33), with 24 cancer-cell and 16 healthy-cell conditions. The cancer panel includes H1299 (non-small-cell lung cancer), A375 (melanoma), Ramos (Burkitt B-cell lymphoma), Jurkat (T-cell leukaemia), MCF-7 (luminal breast adenocarcinoma), and LNCaP (prostate adenocarcinoma). The healthy panel is dominated by MCF-10A (mammary epithelial) and PBMCs (peripheral blood mononuclear cells). The 24 : 16 cancer-to-healthy ratio is the second most favourable of any macro-category in the dataset; furthermore, Paper 12 alone provides 7 H1299 conditions and 15 MCF-10A conditions matched on every parameter (75–105 GHz W-band sweep at 0.2 mW/cm², 1-h exposure), which is by far the best-matched selectivity comparison in the dataset.

D.2 Exposure parameters

Frequency space is split between two regimes. The MMW regime, used by Papers 11, 12 and 32, employs continuous-wave or W-band-swept exposure across 75–110 GHz at 0.2 mW/cm² (a regulator-relevant power density). The THz regime, used by Paper 33, employs continuous-wave 140 GHz exposure at 7.4 mW/cm². Both regimes are unambiguously non-thermal (sub-1 °C temperature rise reported in the source papers). No DC component, no pulsing. The waveform is sine, and within Paper 12 the W-band-sweeping pattern (frequency skipping across 75–105 GHz) is intentional and is presented as a way to probe a broad biological window without committing to any single resonance.

Frequency rationales are predominantly biological-window exploration (the 60–110 GHz region is hypothesised to contain millimetre-scale resonances of biomolecules and water-shell structures), with secondary support from regulatory MMW reference frequencies. The intensity rationale is uniformly sub-thermal regulatory limit, which is the central scientific premise of this macro-category.

D.3 Cell-fate outcomes

The cell-fate readout in macro-category D is dominated by apoptosis (26 conditions) and senescence (22 conditions), with viability secondary (12 conditions). Of the 11 cancer-cell viability measurements, 10 (91 %) reported Direction = -1, the highest cancer-cell viability-reduction rate of any macro-category in this dataset. The single healthy-cell viability measurement reported Direction = -1, but it is from Paper 11 (a paper that contributes only 1 healthy-cell condition) and cannot stand against the much larger and methodologically robust comparison in Paper 12.

In Paper 12 (75–105 GHz W-band sweep, 0.2 mW/cm², 1 h), the apoptosis and senescence readouts are the most striking: H1299 lung cancer cells produced Direction = +1 in 6 of 7 apoptosis conditions, with apoptosis %_vs_control ranging from 332 % to 1542 % (i.e. 3.3-fold to 15-fold induction). Senescence Direction = +1 was reported in 6 of 7 H1299 conditions. In matched MCF-10A controls (15 conditions), every single apoptosis measurement was Direction = 0 and every single senescence measurement was

Direction = 0. This is the cleanest selectivity signal in the entire dataset: under identical exposure parameters, the cancer line undergoes massive apoptosis and senescence induction while the healthy line shows no change in any cell-fate readout.

Necrosis was measured in 4 conditions, Direction mixed (a mix of 0 and +1). Proliferation and motility were not directly measured in this macro-category. Ferroptosis was not measured.

D.4 Mechanistic outcomes

ROS was not measured in macro-category D. Cellular morphology changes were reported in 29 of 40 conditions (mostly in H1299 in Paper 12). Cytoskeleton changes were reported in 6 conditions. Mitochondrial membrane potential was not directly measured. Calcium involvement was not measured. The mechanistic block here is therefore relatively shallow: the cell-fate readout (apoptosis + senescence) is rich, but the upstream mechanistic cause (whether it is ROS, mitochondrial dysfunction, DNA damage, or membrane structural perturbation) is not directly resolved by the dataset.

D.5 Molecular outcomes

Proteomic targets were caspases (Caspase-3, Caspase-9; 4 conditions; Direction = +1, consistent with apoptosis). DNA-damage and apoptosis markers (γ -H2AX, p21) are referenced qualitatively. Genomic data were sparse. The molecular evidence supports a canonical apoptotic execution pathway downstream of an unidentified trigger; γ -H2AX induction would be consistent with DNA double-strand breaks but is not directly quantified in this dataset.

D.6 Mechanism of action emerging within macro-category D

The strongest finding within macro-category D is the W-band MMW selectivity at non-thermal (0.2 mW/cm²) intensity: H1299 → apoptosis + senescence, MCF-10A → no change. Mechanistically, the dataset alone does not resolve whether this is driven by water-structure perturbation, lipid-rafts oscillation, plasma-membrane electrical interface effects, or DNA damage; published literature outside this dataset proposes all four. What can be said with the dataset is that 0.2 mW/cm² in the 75–105 GHz band, applied for 1 h, produces a quantitatively large (3- to 15-fold) apoptotic response selectively in lung cancer cells and a qualitatively coherent senescence response in the same line.

D.7 Limitations of the macro-category D evidence

Two main limitations apply. First, the strongest selectivity result is anchored to a single paper (Paper 12) and a single cell-pair (H1299 vs MCF-10A); whether this generalises to other lung cancers (A549, NCI-H460), other healthy cell types (PBMCs, fibroblasts), or other MMW frequencies is not established within the dataset. Second, the mechanistic chain upstream of apoptosis is not measured: ROS, $\Delta\Psi_m$, calcium, and metabolic readouts are all absent, leaving the “why” of selectivity unresolved. Third, the THz regime (Paper 33) is reported only at the level of cell-line × condition rows but with

apoptosis/senescence/necrosis Direction = 'Not performed', so the comparison MMW vs THz cannot be performed within the dataset.

3.5 Module E — LTE / UHF telecom radiation

E.1 Subgroup overview

Macro-category E comprises 29 conditions from 2 papers (Paper_IDs 31, 35), with 22 cancer-cell and 7 healthy-cell conditions. The cancer panel includes Huh7 and Hep3B (hepatocellular carcinoma), HeLa (cervical adenocarcinoma), SH-SY5Y (neuroblastoma) and B103 (rat neuroblastoma). The healthy panel is ASC (adipose-derived stem cells, Paper 31 and 35) and IMR-90 (human lung fibroblasts, Paper 31). The 22 : 7 ratio is favourable for a selectivity assessment, and the inclusion of two non-cancer cell types in two independent papers strengthens the comparison.

E.2 Exposure parameters

Two carriers are tested: 1.7 GHz (Paper 31) and 1.76 GHz (Paper 35), both LTE / WCDMA modulated (i.e. realistic telecom waveform, not idealised CW). SAR ranges from 0.4 W/kg up to 8 W/kg in Paper 35; Paper 31 uses 1 and 2 W/kg. The 2 W/kg level is the ICNIRP whole-body limit for general public exposure. The waveform rationale is regulatory-relevant: the question being asked is whether realistic mobile-phone-network exposure has biological consequences. Devices are anechoic exposure chambers.

E.3 Cell-fate outcomes

Viability is reported in 13 conditions; of the 12 cancer-cell viability measurements, 0 (zero) reported Direction = -1 — every single cancer-cell viability measurement gave Direction = 0. The single healthy-cell viability measurement also gave Direction = 0. So at SAR ≤ 8 W/kg, LTE exposure does not affect viability over the time-windows tested in either cancer or healthy cells.

Proliferation tells a different story. Of 16 proliferation measurements, 8 of 10 cancer-cell conditions in Paper 31 reported Direction = -1 (a significant reduction in proliferation rate at 1 and 2 W/kg in Huh7, Hep3B, HeLa and SH-SY5Y), but the same paper also reported Direction = -1 in IMR-90 fibroblasts at the same SAR levels — i.e. healthy cells are also affected. ASC was reported with Direction = 0 at 1 W/kg but Direction = -1 at 2 W/kg. In Paper 35 at 1.76 GHz, both cancer (Huh7) and healthy (ASC) cells showed Direction = 0 at all SAR levels (0.4 to 8 W/kg). The dataset therefore does not support a non-thermal LTE-monotheapy hypothesis: the only paper showing a cancer-cell effect (Paper 31) shows the same effect on healthy cells, breaking selectivity.

Apoptosis was measured in 7 conditions: all reported Direction = 0 in cancer cells and 0 in healthy cells in Paper 31; senescence in 4 conditions, also Direction = 0. Motility was not directly measured.

E.4 Mechanistic outcomes

ROS was measured in 4 conditions (cancer): Direction = +1 in all 4, suggesting that LTE exposure does generate oxidative stress in cancer cells even when viability is unchanged. However, this oxidative signal does not translate into an apoptotic readout. Other mechanistic readouts are essentially absent in this macro-category (membrane potential, calcium, cytoskeleton, metabolic = 0 measured conditions).

E.5 Molecular outcomes

DNA-damage and apoptosis-marker proteins (γ -H2AX, cleaved Caspase-3) and MAPK signalling components were profiled in 3+3 conditions, with Direction mostly 0 or weakly positive. The molecular signature does not show a clear cancer-selective derangement.

E.6 Mechanism of action emerging within macro-category E

The dataset suggests that LTE-modulated UHF exposure at sub-thermal SAR can induce a low-grade oxidative stress in cancer cells without translating into apoptosis or selective viability loss. There is no convergent mechanistic chain leading to cancer-selective cell death. The data are best interpreted as biologically real but therapeutically insufficient at this exposure dose.

E.7 Limitations of the macro-category E evidence

The macro-category rests on 2 papers with conflicting findings (Paper 31 reports proliferation effects, Paper 35 reports none), and the mechanistic block is largely empty. There is no clear evidence for selective cancer cytotoxicity at any SAR tested. Higher SAR levels were not explored within this dataset; whether the picture would change at SAR > 8 W/kg, or with longer exposure, cannot be inferred from the present data.

3.6 Module F — AMF intermediate-frequency magnetic field (87 kHz – 320 kHz)

F.1 Subgroup overview

Macro-category F comprises 38 conditions from 3 papers (Paper_IDs 4, 38, 44). It contains 27 cancer-cell conditions and 11 healthy-cell conditions, the latter spanning four healthy lines (NHA — normal human astrocytes; HCF — human cardiac fibroblasts; HUVEC — human umbilical-vein endothelial cells; HDFa — human dermal fibroblasts). The cancer panel is dominated by glioblastoma (U87, LN229, U251, T98, T98G, A172) and includes pancreatic (PANC-1) and breast (MCF-7, MDA-MB-231, MDA-MB-453) lines. Paper 4 alone tests 9 cancer cell lines against 3 healthy lines (NHA, HCF, HUVEC) under matched conditions, which makes this macro-category one of the most informative for selectivity, second only to macro D.

F.2 Exposure parameters

Three carrier frequencies dominate: 227 kHz (Paper 4), 87 kHz and 207 kHz (Paper 38), and 100 kHz (Paper 44). Paper 4 also screens 160, 198, 280 and 320 kHz on U87 and LN229 to identify the optimal frequency. AC magnetic-flux amplitude is 18 mT (Paper 4), 1.4–1.6 mT (Paper 38) and 10 mT (Paper 44). Exposure duration is 30 min (Paper 4), 60 min (Paper 38) and 60–90 min (Paper 44). All exposures are sinusoidal. The intensity rationale is engineering-driven (achieving sufficient induced eddy currents within the cells without raising bulk medium temperature) and the frequency rationale is empirical optimisation. Devices are custom-built solenoids or Helmholtz coils with active temperature control to ensure non-thermal regime.

F.3 Cell-fate outcomes

Viability is the dominant readout: 38 conditions (cancer = 27, healthy = 11). Of the 27 cancer measurements, 15 (56 %) reported Direction = -1, 12 reported Direction = 0, 0 reported +1. Of the 11 healthy measurements, 0 reported Direction = -1, 11 reported Direction = 0, 0 reported +1. This is one of the cleanest selectivity profiles in the dataset on viability: cancer-line damage in over half the conditions vs zero damage to any healthy line tested.

Within Paper 4 (227 kHz, 18 mT, 30 min, observation 24 h post), the cancer-cell viability reductions in the most sensitive lines were striking: U87 to 60 % of control, A172 to 48 %, MCF-7 to 48 %, MDA-MB-231 to 71 %, MDA-MB-453 to 77 %, LN229 to 59 %, U251 to 70 %, T98 to 80 %, PANC-1 to 67 %. Across the same paper, the three healthy lines (NHA = 90 %, HCF = 105 %, HUVEC = 96 %) all gave Direction = 0. The frequency screen on U87 (160, 198, 227, 280, 320 kHz) showed a clear maximum sensitivity at 198–227 kHz, with effects fading at the band edges (320 kHz = 92 % of control, Direction = 0). This is the strongest dose–frequency response in this macro-category and one of the strongest in the entire dataset.

Paper 38 (87 and 207 kHz, ~1.5 mT) tested SH-SY5Y (cancer), T98G (cancer), NHA (healthy) and HDFa (healthy). Effects were small and most conditions reported Direction = 0; only one SH-SY5Y condition at 207 kHz reached Direction = -1 (66 % of control after 1 h exposure / 30 h observation). Paper 44 reported

large reductions (51–82 %) in MDA-MB-231 at 100 kHz / 10 mT but did not include matched healthy controls. The dose–response across these three papers therefore shows a clear amplitude threshold: at ~1.5 mT effects are weak; at 10–18 mT effects are large; the magnetic-field amplitude is the dominant predictor of efficacy in this band.

Apoptosis was measured in 8 conditions (cancer 4, healthy 4): in cancer cells all 4 measurements reported Direction = 0; in healthy cells all 4 also Direction = 0. So the cell-fate route is not classical apoptosis at the time-points sampled. Necrosis was measured in 8 conditions (cancer 4, healthy 4) — Direction values mixed, but no consistent pattern. Senescence and motility were not measured. Ferroptosis was not measured.

F.4 Mechanistic outcomes

The mechanistic block in macro-category F is the most informative on antioxidant response. Antioxidant_Direction was measured in 9 conditions; Direction = -1 was the dominant pattern in cancer cells (Paper 4), consistent with depletion of GSH or down-regulation of antioxidant enzymes. ROS_Direction was Direction = +1 in 2 of 3 cancer conditions and 0 in the third. Cellular morphology changes were reported in 9 conditions, cytoskeleton in 9 conditions, metabolic reprogramming in 3 conditions. Mitochondrial membrane potential was measured in only 1 condition (Direction = +1), preventing strong conclusions on $\Delta\Psi_m$. Cell cycle was measured in 10 conditions, of which Paper 38 phase-resolved 2 conditions in SH-SY5Y (G0/G1 arrest signature).

F.5 Molecular outcomes

Genomic targets were DNA-integrity probes (γ -H2AX foci, comet assay endpoints; 8 conditions) and metabolic-pathway transcripts (2 conditions). Proteomic targets were glycolysis enzymes (LDH-A, HK2, PKM2), mitochondrial-biogenesis markers (PGC-1 α), and cell-cycle regulators (cyclin D1, p21) — 3 conditions in Paper 4. The molecular profile is consistent with a metabolic-stress mechanism: a 30-min 227 kHz / 18 mT exposure appears to depolarise mitochondria (suggested by antioxidant depletion and ROS rise), shift glycolytic flux, and trigger cell-cycle arrest in G0/G1, leading to a non-apoptotic loss of viability over 24 h.

F.6 Mechanism of action emerging within macro-category F

Synthesising the four blocks above, the proposed mechanism in the strongest-effect condition (Paper 4, 227 kHz, 18 mT, 30 min, 24 h observation) is: the alternating magnetic field induces eddy currents inside the cytoplasm with a current density that scales as $f \times B \times (\text{cell radius})^2 \rightarrow$ the induced currents preferentially load mitochondrial membranes that are already depolarised in cancer cells (Warburg phenotype: mitochondria with elevated $\Delta\Psi_m$ and dysregulated respiratory chain) $\rightarrow$ mitochondrial perturbation generates a ROS pulse, depletes the antioxidant pool, and triggers a metabolic-arrest signal (cyclin D1 $\downarrow$, p21 $\uparrow$) $\rightarrow$ cancer cells fail to traverse G1/S; healthy cells, with their lower mitochondrial-membrane charge density and intact antioxidant reserve, buffer the same induced-current loading and

recover. This mechanism is consistent with the paper-4 finding that 9 cancer cell lines respond and 3 healthy lines do not.

F.7 Limitations of the macro-category F evidence

The strongest selectivity result is anchored to a single paper (Paper 4); replication across an independent group at 227 kHz / 18 mT is not present in the dataset. The dose–response curve of frequency on U87 / LN229 (Paper 4) is convincing but is internal to one experimental setup. Apoptosis was uniformly Direction = 0 in this category, which is biologically interesting (the death route is non-apoptotic) but means the molecular execution machinery is unresolved. The link between the ROS/antioxidant readout and the viability readout is plausible but not directly demonstrated by, e.g., antioxidant rescue experiments within this dataset.

3.7 Module G — RF MHz fields (radiofrequency, AC + DC bias)

G.1 Subgroup overview

Macro-category G comprises 30 conditions from 4 papers (Paper_IDs 7, 29, 30, 37), all on cancer cells and zero on healthy cells. The cancer panel is HT-1080 (fibrosarcoma), MCF-7 (breast adenocarcinoma) and three colorectal lines (HT29, SW480, LoVo). The complete absence of healthy-cell conditions is a fundamental limitation that prevents any direct selectivity assessment within this macro-category.

G.2 Exposure parameters

The defining feature of this macro-category is the combination of a sinusoidal AC magnetic field in the MHz range (1.8, 3.6, 4.5, 7.2 MHz, and 13.56 MHz) with a static DC magnetic field that is many times larger in amplitude (1–300 μT DC vs 1 nT–20 nT AC, in Papers 7 and 29). In some Paper 29 conditions the DC amplitude is set explicitly to the local geomagnetic value ($\sim 45 \mu\text{T}$) to test resonance against natural magnetic conditions; in others it is reduced to 10 μT or raised to 90–300 μT . Paper 30 reports neither AC frequency nor amplitude (test of an ambient-equivalent field). Paper 37 (REAC at 13.56 MHz) tests a clinical RF device at unspecified amplitude.

The frequency rationale for this macro-category is biophysically interesting: most of these exposures are designed to probe Larmor-resonance conditions for biologically relevant ions (Ca^{2+} , K^+ , Na^+ , Mg^{2+}) by tuning the AC frequency against the static DC bias according to $f_{\text{Larmor}} = (\gamma/2\pi) \times B_{\text{DC}}$. This places the macro-category in a different mechanistic space from any other in the dataset.

G.3 Cell-fate outcomes

Viability was measured in 10 conditions: 6 reported Direction = -1, 1 Direction = 0, 3 Direction = +1. The strongest reductions came from Paper 7 at 2 MHz / 20 nT AC / 45 μT DC (HT-1080 viability 96 % of control, Direction = -1) and Paper 30 (MCF-7 viability 84.5 % of control, Direction = -1). The +1 measurements came from Paper 7 at 2.5, 4 and 4.5 MHz under the same DC bias, indicating a sharp resonance-like alternation between cytotoxic and pro-proliferative effects across a 2.5-MHz frequency window. This bidirectional pattern is unique within the dataset and is biophysically consistent with an ion-cyclotron-resonance hypothesis.

Proliferation was measured in 3 conditions, all Direction = +1 in Paper 7 (a pro-proliferative response in HT-1080 at 4 and 4.5 MHz). This is a counterproductive finding from a therapeutic standpoint: certain RF-MHz / static-bias combinations stimulate cancer-cell growth rather than suppressing it. Apoptosis was Direction = +1 in 5 of 6 conditions where it was measured, but only at the cytotoxic frequencies (2 and 3.5 MHz), confirming that the mechanism behind the -1 viability is apoptotic. Motility was Direction = -1 in all 3 conditions where it was measured (cancer).

G.4 Mechanistic outcomes

ROS was measured in 16 conditions (the highest ROS coverage of any macro-category in the dataset). Direction = +1 in 2 conditions, Direction = 0 in 5, Direction = -1 in 9. The dominance of Direction = -1 (a reduction in ROS in cancer cells under RF exposure) is biologically counter-therapeutic relative to the success criterion (we want ROS_Direction = +1 in cancer); 2 of those Direction = -1 conditions are at the cytotoxic frequencies (3.6 MHz, 1 nT), suggesting that the kill mechanism in macro-category G is not classical oxidative stress but more likely calcium-channel disruption.

Calcium involvement was probed in 14 conditions (Direction = +1 in 2, Direction = 0 in 6, Mentioned in 6) — the highest direct calcium coverage in the dataset. This is consistent with the ion-cyclotron-resonance design intent. Mitochondrial membrane potential was measured in 10 conditions, with Direction split (4 of +1, 4 of 0, 2 of -1). The mechanistic picture in macro-category G is therefore a calcium-and-mitochondrial story rather than a ROS story.

G.5 Molecular outcomes

Genomic targets included stemness markers (Oct4, Sox2), cell-cycle / senescence regulators (p16, p21), and apoptosis / stress-response genes (Bax, Bcl-2, p53), with 3 conditions reporting Direction = mixed depending on frequency. Proteomic targets focused on secreted tumour biomarkers and autophagy markers (LC3-II, Beclin-1, p62), with 3 conditions reporting Direction = +1, consistent with autophagy induction.

G.6 Mechanism of action emerging within macro-category G

The proposed mechanism is calcium-cyclotron-resonance modulation: under specific AC/DC combinations, the alternating field couples to specific ion-cyclotron transitions, producing a calcium-flux pattern that the cancer cells cannot buffer; this in turn perturbs mitochondrial Ca^{2+} uptake, depolarises mitochondria, triggers autophagy, and at the right frequencies leads to apoptosis. At off-resonance frequencies, the calcium pattern instead drives proliferative signalling (the +1 viability and +1 proliferation observed at 2.5, 4 and 4.5 MHz). The mechanism is biophysically attractive and would provide a sharp frequency-tuning lever for therapy, but the absence of healthy-cell data within this macro-category leaves the selectivity claim unsupported.

G.7 Limitations of the macro-category G evidence

Three limitations are decisive. First, no healthy-cell condition is included, so the apparently selective mechanism cannot be tested for selectivity within the dataset. Second, several conditions in Paper 7 report a pro-proliferative effect, which is a serious therapeutic risk that must be characterised before any clinical translation. Third, Paper 30 lacks frequency and amplitude reporting, which limits the inferences that can be drawn from its 3 conditions.

3.8 Module H — Static magnetic field (SMF)

H.1 Subgroup overview

Macro-category H comprises 4 conditions from 2 papers (Paper_IDs 18 and 44), all on cancer cells (HT-1080 and MDA-MB-231). The evidence base is too small to support population-level inference; the macro-category is included for completeness and as a negative control for the AC-bearing macro-categories (F, G, I).

H.2 Exposure parameters

Paper 18 used a -45 μ T bias (i.e. partial cancellation of the local geomagnetic field). Paper 44 used a 15 mT static field. The two intensities differ by three orders of magnitude. Frequency is N/A by definition; only the DC magnetic-flux density and exposure duration matter.

H.3 Cell-fate outcomes

Viability was measured in 2 conditions (Paper 44 only): both reported Direction = -1 (MDA-MB-231 reduced under 15 mT static exposure). Proliferation was measured in 2 conditions (Paper 18, HT-1080): one Direction = -1 and one Direction = +1, illustrating that even within one paper at one DC value the sign of the proliferative response is unstable.

H.4 Mechanistic outcomes

Membrane potential was measured in 1 condition (Direction = -1) and antioxidant in 1 condition. Calcium and cytoskeleton were each measured in 1 condition. Coverage is too low for any synthesis.

H.5 Molecular outcomes

No genomic or proteomic data reported.

H.6–H.7 Mechanism and limitations

The mechanistic narrative cannot be supported with 4 conditions from 2 papers. The macro-category should be interpreted as motivating, not establishing, that very strong static fields (15 mT scale) may cytotoxically affect TNBC cells, and that cancellation of the geomagnetic field may modulate fibrosarcoma proliferation in either direction. Additional papers and matched healthy controls would be required.

3.9 Module I — ELF-EMF combined with SMF

I.1 Subgroup overview

Macro-category I comprises 12 conditions from 3 papers (Paper_IDs 9, 18, 36), with 9 cancer-cell and 3 healthy-cell conditions. Cancer lines are G401 (Wilms tumour), A549 (lung adenocarcinoma), HT-1080 (fibrosarcoma), MDA-MB-231 (TNBC) and HT-29 (colorectal). Healthy cells are HEK293 (Paper 9) and HDF (human dermal fibroblast, Paper 18). The 9 : 3 ratio allows limited selectivity assessment but the heterogeneity of conditions across the 3 papers makes per-paper selectivity assessment more reliable than aggregate.

I.2 Exposure parameters

Paper 9 uses a 50 Hz sinusoidal AC field at 1.0–2.5 mT superimposed on a 3.0–4.0 mT DC field (a strong combined exposure). Paper 18 uses ELF in a Larmor-resonance regime at 14.5–16.5 Hz, AC at 10–18 μ T, DC at 45 μ T (geomagnetic). Paper 36 uses 6 Hz and 24 Hz AC at 100 μ T with 45 μ T DC bias. The three papers therefore probe three quite different biophysical regimes within the same macro-category, which is itself an interesting comparative element.

I.3 Cell-fate outcomes

Proliferation was the dominant readout (11 of 12 conditions). In Paper 9 (50 Hz, 1–2.5 mT AC + 3–4 mT DC), G401 and A549 both showed Direction = -1 (cancer) and HEK293 also showed Direction = -1 — i.e. selectivity is broken in this paper, the field affects healthy cells too. In Paper 18, HT-1080 showed Direction = +1 (pro-proliferative) at 14.5–16.5 Hz / 10–18 μ T, except for one condition at 16.5 Hz / 18.5 μ T where Direction flipped to -1; HDF healthy fibroblasts showed Direction = 0 in both 14.5 Hz conditions tested. In Paper 36, MDA-MB-231 (6 Hz) and HT-29 (24 Hz) both showed Direction = -1. The cleanest selectivity case in macro-category I is therefore Paper 18 at 16.5 Hz / 18.5 μ T, where a small region of parameter space gave -1 in cancer with 0 in healthy, but it is a single condition.

Apoptosis was measured in 2 conditions (cancer): both Direction = +1, in Paper 9. Motility was measured in 2 conditions: both Direction = -1 in cancer. ROS was measured in 5 conditions (Direction = +1 in 2, 0 in 2, +1 in 1). Antioxidant in 2 conditions.

I.4 Mechanistic outcomes

Calcium involvement is the standout mechanism in this macro-category, with 5 conditions reporting it (Direction = +1 in 1, 0 in 1, Mentioned in 2, Not performed in 1). Ferroptosis was measured in 2 conditions (Paper 9), Direction = +1 — the only two flagged ferroptosis +1 conditions in the entire dataset. Mitochondrial membrane potential was measured in 1 condition (+1). Metabolic reprogramming was measured in 2 conditions, cytoskeleton in 2.

I.5 Molecular outcomes

DNA damage repair genes and proteins (γ -H2AX, BRCA1, RAD51) were the dominant molecular targets, in 2+2 conditions, with Direction mixed. The molecular profile is consistent with a DNA-damage / repair perturbation, which dovetails with the ferroptosis flag in Paper 9 (lipid peroxidation as a downstream consequence).

I.6 Mechanism of action emerging within macro-category I

The proposed mechanism is resonance-tuned ion-flux modulation: the AC/DC combination exploits Larmor or cyclotron resonance for specific ions (Ca^{2+} in particular), producing a transmembrane ion-flux pattern that, combined with DNA-repair perturbation, pushes cancer cells toward apoptosis or ferroptosis. The most therapeutically interesting condition is Paper 18 / 16.5 Hz / 18.5 μT , but it is single. Paper 9 shows that at higher AC/DC amplitudes the selectivity is lost (HEK293 also dies).

I.7 Limitations of the macro-category I evidence

The principal limitation is parametric heterogeneity: 3 papers, 3 fundamentally different (AC, DC, frequency) regimes, 12 conditions. There is no replication across papers at any single (frequency, AC, DC) point. The ferroptosis +1 finding is restricted to 2 conditions in 1 paper and would benefit from replication.

3.10 Module J — Rotating / spinning / gradient ELF magnetic fields

J.1 Subgroup overview

Macro-category J comprises 20 conditions from 3 papers (Paper_IDs 8, 23, 40), with 18 cancer-cell and 2 healthy-cell conditions. Cancer lines are GBM115 and DIPG (paediatric high-grade glioma; Paper 8), MDA-MB-231, BT549, MDA-MB-468 (Paper 23) and MCF-7 (Paper 40). Healthy partners are MCF-10A and RPE1 (retinal pigment epithelium, Paper 40).

J.2 Exposure parameters

The defining feature of this macro-category is spatial inhomogeneity (rotating, spinning, or gradient) of the magnetic field at low frequency. Paper 8 uses a spinning oscillating magnetic field at 137 and 272 Hz, 5 mT amplitude; Paper 23 uses a gradient rotating magnetic field at 2.5 and 5 Hz, 0.04–0.41 T (very high amplitude); Paper 40 uses LF-RMF at 4.2 Hz, 0.1–0.4 T. The amplitudes are at the upper end of what is physically achievable in a benchtop magnet and are far above ELF-general μ T-scale conditions.

J.3 Cell-fate outcomes

Viability was measured in 5 conditions (cancer only): all 5 reported Direction = -1, with the strongest reductions at 137 Hz / 5 mT in DIPG (20 % of control) and GBM115 (40 % of control) in Paper 8. Motility was measured in 8 conditions, all cancer, all Direction = -1 — i.e. every motility measurement in this macro-category showed a significant inhibition of migration. This is the strongest motility signal in the dataset and is the unique selling-point of macro-category J.

Proliferation was measured in 1 condition (Direction = -1) and apoptosis in 2 conditions (both Direction = +1). The 2 healthy-cell conditions (MCF-10A and RPE1, Paper 40) reported Direction = Not performed for viability and motility, so a direct selectivity assessment in this macro-category is restricted to descriptive comparison rather than matched-condition analysis.

J.4 Mechanistic outcomes

Cellular morphology was reported in 12 conditions, cytoskeleton in 12 conditions — the most cytoskeleton-rich coverage of any macro-category proportionally. ROS was measured in 4 conditions (2 of +1, 2 of 0). Membrane potential was measured in 1 condition (Direction = 0). Metabolic reprogramming in 6 conditions. The cytoskeleton dominance is biologically expected: a rotating or strongly-gradient magnetic field is hypothesised to apply mechanical torques on diamagnetic cell components (microtubules, actin filaments, mitochondria themselves), producing cytoskeletal disorganisation that translates into the observed motility loss.

J.5 Molecular outcomes

The proteomic targets are exactly what the mechanism predicts: Rho GTPases (RhoA, Rac1, Cdc42; 2+2 conditions, Paper 23), focal-adhesion proteins (FAK, paxillin), and TGF- β 1/SMAD3-axis EMT markers (1 condition, Paper 40, all Direction consistent with EMT inhibition).

J.6 Mechanism of action emerging within macro-category J

The synthesised mechanism is mechanical torque on diamagnetic cytoskeletal elements: the rotating or spatially-gradient magnetic field exerts forces on microtubules and actin (and possibly on mitochondrial cristae), producing cytoskeletal disorganisation, focal-adhesion remodelling (Rho GTPase down-regulation), and consequent loss of polarised migration. This mechanism predicts exactly the motility-dominated readout observed (8/8 Direction = -1) and is internally consistent across the 3 papers despite their different field geometries.

J.7 Limitations of the macro-category J evidence

The principal limitation is the absence of viability/motility readouts in the healthy-cell partners (MCF-10A, RPE1 in Paper 40), which prevents a direct selectivity assessment of the motility inhibition. Whether MCF-10A migration is also inhibited at the same field parameters cannot be answered from this dataset. The very high field amplitudes (0.1–0.4 T) make benchtop translation to clinical settings non-trivial.

3.11 Module K — ELF-EMF general (sinusoidal & pulsed, single-axis): rationale for sub-partitioning

Macro K is the largest group of the dataset (148 conditions, 18 papers) and the most parametrically heterogeneous: frequencies span 0.01 Hz to 80 Hz, amplitudes span 1 μ T to >2 T, and waveforms include pure sinusoid, square pulse, multifrequency pulses, and complex patterned signals. Aggregating these biophysically incompatible regimes obscures rather than clarifies. I therefore partitioned macro K, by waveform and amplitude regime, into three sub-modules: K1 — pure sinusoidal ELF at physiological amplitudes (≤ 5 mT); K2 — pulsed/square ELF at higher amplitudes (≥ 20 mT typical, up to >2 T); K3 — under-specified or complex waveforms (Wave_Type 'Not reported', 'Sine-like', or complex patterned). The three sub-modules are mutually exclusive and jointly cover all 148 K-rows. Their differential signatures are reported below.

Sub-module K1 contains 52 rows from 7 papers (P2, P3, P13, P19, P28, P41, P48) with a 42 cancer / 10 healthy split. K2 contains 68 rows from 5 papers (P1, P5, P10, P14, P20) with a 42 cancer / 26 healthy split — the only macro-K sub-module with substantial healthy-cell coverage. K3 contains 28 rows from 6 papers (P6, P15, P16, P21, P24, P25) with a 22 cancer / 6 healthy split.

3.12 Module K1 — Sinusoidal ELF at physiological amplitudes (≤ 5 mT)

K1.1 Subgroup overview

K1 contains 52 conditions from 7 papers, all using a pure sinusoidal waveform (Wave_Type = 'Sine') at amplitudes spanning 0.1 mT (Paper 19) to 35 mT (Paper 3) but with the modal regime at 0.1–4.5 mT, well within or only slightly above occupational reference levels. Cell lines include MCF-7, MDA-MB-231 and MCF-10A (the canonical breast triad in Papers 19 and 41), DU-145 prostate (P3, P13), HEK293 with 786-O / 769-P / Caki1 renal-cancer panel (P28), HeLa/CaSki/SiHa cervical-cancer panel co-cultured with primary CD8⁺ T cells (P48), PANC-1 / SKBR3 / 1627 (P2), HUVEC and K562 controls (P13). The sub-module is the most pertinent to occupational-exposure science and to the 'biological-windows' literature.

K1.2 Exposure parameters

Frequency is dominated by 50 Hz (29 conditions, the European power-line frequency) and 20 Hz (5 conditions, P48). Lower-frequency probes (3, 6, 8, 10, 14 Hz) appear in P2's biological-window scan; 60 Hz appears once. AC amplitudes cluster at 0.1 mT (P19), 1 mT (P13, P19, P41), 4.5 mT (P28), 22.6 / 35 mT (P3), and 70 μ T (P2). Exposure durations span 1 h to 21 days, with 24–72 h being the modal regime. Frequency rationales are split between bioactive-window exploration and power-line-frequency regulatory relevance; intensity rationales follow occupational reference levels (ICNIRP).

K1.3 Cell-fate outcomes

Viability was reported in 33 conditions (the rest 'Not performed'). In cancer cells (24 conditions reported with a Direction value), 13 reported Direction = -1 (54 %), 9 reported +1 (37 %), and 2 reported 0. The high +1 fraction is the headline-counter-therapeutic signal of K1 and is concentrated in P19 (MCF-7 and MDA-MB-231 at 50 Hz / 0.1 mT, hormetic increases to 120–127 % of control) and P21 (the +1 conditions discussed under K3) — i.e. at the lowest amplitudes the response inverts. In healthy cells (9 reported), 2 are -1, 4 are 0, and 3 are +1, so the healthy harm rate within K1 is 22 % (2/9). Selectivity index for K1 = $0.54 / \max(0.22, 0.05) \approx 1.32$ (Figure 1) — modest at the aggregate level.

Proliferation was reported in 35 conditions (29 cancer, 6 healthy with directions). Cancer Direction = -1 in 16/29 (55 %), +1 in 4/29 (14 %), 0 in 9/29 (31 %). Healthy Direction = -1 in 4/6 — i.e. healthy cells in K1 are reduced in proliferation in two-thirds of measured conditions, undermining the apparent cancer-anti-proliferative signal. Apoptosis/necrosis/senescence are sparsely reported (16 / 8 / 0 cancer conditions respectively). Motility is reported in 7 cancer conditions: 5 are Direction = -1 (Paper 28 in renal-cancer lines, suggesting a possible anti-migratory window at 50 Hz / 4.5 mT).

Paper 19 deserves special highlighting because it provides the cleanest amplitude-titration in the entire dataset: 50 Hz delivered at 0.1 mT vs 1 mT in MCF-7, MDA-MB-231 and MCF-10A. At 0.1 mT, MCF-7 and MDA-MB-231 viabilities are 120–127 % of control (Direction +1, hormetic stimulation) and MCF-10A is at Direction = 0. At 1 mT, MCF-7 drops to 82 % and MDA-MB-231 to 85 % of control (Direction -1) while MCF-10A is at 80 % (Direction -1) — i.e. selectivity is lost at the higher amplitude and the field becomes nonspecifically cytotoxic. Within K1 there is therefore no obvious therapeutic window; K1 is more compatible with adjuvant or sensitiser logic than with monotherapy.

K1.4 Mechanistic outcomes

$\Delta\Psi_m$ was reported in 12 K1 cancer conditions (8 with Direction): 3 Direction = -1, 4 = 0, 1 = +1. ROS reported in 23 K1 cancer conditions (10 with Direction): 7 = +1, 3 = 0. The combined $\Delta\Psi_m\downarrow$ /ROS $\uparrow$ chain is therefore present but not dominant in K1, consistent with sub-mitochondrial-stress amplitudes. Cell-cycle was reported in 8 conditions, with G0/G1 arrest dominant in P28 (4.5 mT/50 Hz on renal-cancer lines).

K1.5 Molecular outcomes

Genomic data are sparse in K1 (one condition with Direction). Proteomic data come from P28 (3 conditions, differential proteomics in 786-O / 769-P / Caki1) reporting modulation of focal-adhesion, mitochondrial, ribosomal and RNP-complex protein groups, consistent with a metabolic-and-cytoskeletal readjustment phenotype rather than a frank pro-apoptotic one.

K1.6 Mechanism of action emerging within K1

K1 evidence supports a hormetic biphasic model: at 0.1 mT / 50 Hz, sinusoidal ELF stimulates mitochondrial biogenesis and antioxidant capacity, mildly favouring proliferation in both cancer and healthy lines; above ≈ 1 mT at 50 Hz, the same machinery transitions to a low-grade stress response ($\Delta\Psi_m$

↓, ROS ↑, p21 ↑, G0/G1 arrest), with cytotoxicity that is not selective for cancer over isogenic non-tumorigenic controls. This is the dominant operating regime for occupational-exposure science but it is the wrong regime for monotherapy.

K1.7 Limitations of K1 evidence

Healthy-cell coverage is small (9 conditions, 3 isogenic pairings: MCF-10A vs MCF-7/MDA-MB-231; HEK293 vs renal panel; CD8⁺ T cells vs cervical lines). The 0.1 mT pro-proliferative findings are concentrated in one paper (P19) and have not been independently replicated in this dataset. Apoptosis / motility / cell-cycle are under-sampled. The biological-window claim of P2 (low-Hz frequency scan in PANC-1) rests on a single paper at 70 μT and should not be over-interpreted.

3.13 Module K2 — Pulsed/square ELF at higher amplitudes

K2.1 Subgroup overview

K2 contains 68 conditions from 5 papers (P1, P5, P10, P14, P20) with a 42 cancer / 26 healthy split — the most healthy-cell-rich sub-module of macro K and one of the most rigorous in the entire dataset. Cell lines include CT2A glioma + N2A neuroblastoma (P1) with isogenic C8D1A astrocytes; U373 MG glioma + HeLa + MC-38-Luc colorectal with HaCaT keratinocytes + CV-1 + COS-1 monkey-kidney controls (P5); MDA-MB-231 + MCF-7 with FF95 fibroblasts (P10); B16F10 melanoma + SKBR3 + MDA-MB-231 + PC12 with NIH/3T3 + 3T3 (P14); HeLa with NIH/3T3 (P20). All papers report Wave_Type = 'Square pulse' or 'Pulsed (multifrequency)'.

K2.2 Exposure parameters

Frequency is dominated by 45–55 Hz (the band that brackets European/US power-line frequencies, 32 conditions across P1 and P14), with 7.83 Hz (Schumann resonance, P20), 8 Hz (P10), 30–80 Hz (P14), and a P5 multifrequency pulsed exposure (<30 Hz) labelled 'Pulsed (multifrequency)'. AC amplitudes cluster at 100 μT (P1, P14), 11 μT (P10), 200 / 600 μT (P20), and a remarkable >2 T super-strong pulsed regime in P5 (ms-pulse single-shot). Exposure durations span 1 h to 72 h. The Crocetti rationale (Paper 1) — that the bioactive frequency is set by the thermal-Boltzmann-vs-biomolecular-dynamics crossover — is the most theoretically articulated rationale in the entire dataset.

K2.3 Cell-fate outcomes

Viability was reported in 62 K2 conditions, including 41 with Direction values (and 21 'Not performed'). In cancer cells (39 with directions), 36 reported Direction = -1 (92 %, the highest cancer-cytotoxicity rate of any macro-K sub-module), 3 reported +1, and 0 reported 0. In healthy cells (23 with directions), 11 reported -1 (48 %), 7 reported 0, and 5 reported +1. So K2 is highly cytotoxic to cancer but also damages healthy cells in roughly half the measured conditions; the selectivity index is approximately 0.92 / 0.48 ≈ 1.92 (Figure 1) — significantly worse than F, C, J or D and similar to TTFields without their dose-window selectivity.

Paper-level diagnosis: P1 (Crocetti, 100 μ T 24–72 h) reduces CT2A glioma viability to 58–80 % of control (8/8 Direction = -1) and N2A neuroblastoma to 63–90 % (9/9 -1), but C8D1A astrocyte viability is also reduced to 71–87 % at 30/45/50/55 Hz (5/9 healthy -1), eroding selectivity; only at 20/40 Hz are astrocytes spared (Direction +1, mild stimulation). P14 confirms NIH/3T3 reduction at 45–50 Hz / 100 μ T (3T3: 80, 88 %, Direction -1), reinforcing the non-selectivity of the 45–55 Hz / 100 μ T regime. P5 (super-strong ms-pulse) reduces U373 MG, HeLa and MC-38-Luc strongly but also damages HaCaT, CV-1 and COS-1, with healthy harm essentially equal in magnitude — i.e. above the magnetostimulation threshold the field is non-specifically cytotoxic.

Proliferation was reported in 51 K2 conditions, with 20 cancer Direction = -1 (49 %), 7 +1, 1 = 0. Healthy proliferation: 3 -1, 11 = 0, 3 +1 (the two highest values are P14 NIH/3T3 at 80 Hz, where the field has no proliferation effect on the fibroblast). Apoptosis was reported in 18 conditions, 5 cancer Direction = +1 (P1, P14). Necrosis Direction values were 0/0 — not observed.

K2.4 Mechanistic outcomes

Cell cycle is the most-measured mechanistic block in K2: 11 conditions with G0/G1 arrest in P1 and P14, consistent with mitochondrial-stress-driven cycle disruption. $\Delta\Psi$ m and ROS were not directly measured in K2 (P1 reports 'metabolic shift' as a derived endpoint; P5 mentions ROS but does not report Direction). Calcium / morphology / cytoskeleton flags are positive in 20+ K2 conditions but with thin Direction reporting.

K2.5 Molecular outcomes

Mitochondrial / respiratory-chain genes (NDUFA, COX-IV, ATP synthase subunits) and cell-cycle regulators (CDKN1A/p21, cyclin D1) dominate the K2 genomic block (P1, P14), with directions consistent with mitochondrial-stress $\rightarrow$ p21 $\uparrow$ $\rightarrow$ cyclin D1 $\downarrow$ $\rightarrow$ G0/G1 arrest. Pluripotency factors (Sox2, Oct4) are down-regulated in CT2A glioma (P1), supporting a de-stemming hypothesis. P5's super-strong ms-pulse adds RNA-stability / heat-shock signatures.

K2.6 Mechanism of action emerging within K2

K2 evidence supports a bona-fide mitochondrial-stress / G0/G1 arrest mechanism for ELF-EMF cancer cytotoxicity, but operating via a non-selective biophysical channel — both glioma and astrocytes, both melanoma and fibroblasts, are reduced at the same exposure parameters. Translationally, K2 is therefore a strong candidate for an adjuvant / sensitiser role (where the host healthy tissue is already protected by normal repair) but a poor candidate for monotherapy. The 20–40 Hz / 100 μ T window (P1) where astrocytes are spared but glioma is damaged is the only candidate selective window, and rests on a single paper.

K2.7 Limitations of K2 evidence

All 5 K2 papers use square / pulsed waveforms at amplitudes $\geq 10 \mu\text{T}$; the mechanism appears to be insensitive to the precise pulse shape. The high-amplitude regime (P5) and the biological-window regime (P1) are mechanistically different but lumped into K2 here for waveform homogeneity; a future iteration could split K2a (low-amplitude pulsed, P1/P10/P14/P20) from K2b (super-strong pulsed, P5). Selectivity-window claims rest on P1 alone for the 20–40 Hz / 100 μT case.

3.14 Module K3 — Specialty / under-specified ELF waveforms

K3.1 Subgroup overview

K3 contains 28 conditions from 6 papers (P6, P15, P16, P21, P24, P25) with a 22 cancer / 6 healthy split. The K3 conditions share a feature absent in K1 and K2: the waveform is either explicitly complex (P21 — 'Complex patterned, modeled after intercellular calcium-wave firing pattern'), under-specified ('Not reported' in P6, P15, P19's cousin block, P24, P25), or 'Sine-like (not explicitly specified)' in P16. Cell lines include U87 + U251 + U251-P53 glioma (P6) with normal human fibroblast and MCF-7 cross-control; HT-1197 bladder cancer (P15); 293T + HepG2 + A549 + Raji (P16); B16-BL6 melanoma + HEK293T (P21, calcium-wave-pattern conditioning); A549 + LLC + MDA-MB-231 (P24); MC4-L2 mammary tumour line (P25).

K3.2 Exposure parameters

Frequency includes 1 Hz (P6, P13, P25, three independent papers, 100 mT each), 30 Hz (P15, P14 pair), 20 Hz (P16 at 5 mT), 50 Hz + 385 Hz combined (P24), and 7.83 Hz (P21, complex patterned at 1 μT). Amplitudes span 1 μT (P21) to 100 mT (P6, P13, P25). The wave types are heterogeneous, with the calcium-wave-pattern conditioning (P21) representing the most theoretically distinctive entry.

K3.3 Cell-fate outcomes

Viability was reported in 12 K3 conditions: 7 cancer Direction = -1 (78 %), 0 cancer +1, 2 cancer 0. Healthy viability: 0 -1, 3 = 0 (P6 normal human fibroblast, P21 HEK293T, P24 hLEC) — i.e. zero healthy cytotoxicity in the 6 K3 healthy conditions reported. Selectivity index for K3 $\approx 0.78 / 0.05$ (floor) = 15.6 (capped at 4.20 in Figure 1 because of the small healthy-n caveat). This is by far the cleanest selectivity signal within macro K, and rivals macros C, F and J at the aggregate level.

Driver papers: P6 (1 Hz / 100 mT for 24–72 h on U87/U251/U251-P53) reduces glioma viability to 70–85 % of control (3/3 -1), spares normal human fibroblast (Direction = 0); P21 calcium-wave-pattern at 1 μT reduces B16-BL6 melanoma in 1/2 conditions but spares HEK293T (Direction = 0); P25 reduces MC4-L2 mammary tumour viability at 1 Hz / 100 mT (Direction = -1, no healthy comparator). The convergence of three independent papers at 1 Hz / 100 mT on a Direction = -1 cancer-cytotoxicity reading is the strongest emergent finding of K3 and should be highlighted as a candidate biophysical window distinct from the 50 Hz canonical regime.

Proliferation was reported in 20 K3 conditions, with 11 cancer Direction = -1 (52 %), 5 = 0, 0 +1. Necrosis is reported at 36 % (Direction values reported in 10 conditions, the highest necrosis-reporting rate of macro K). Apoptosis is reported in 6 cancer conditions (29 % Direction +1).

K3.4 Mechanistic outcomes

$\Delta\Psi_m$ was reported in 12 K3 cancer conditions, with 4 Direction = -1 — the highest $\Delta\Psi_m$ coverage rate within macro K. Cytoskeletal disruption was flagged (P6, P15, P21). ROS was reported in only 1 K3 condition.

K3.5 Molecular outcomes

Molecular data within K3 are sparse but interesting: P6 reports modulation of glioma stem-cell markers (Sox2 down) under 1 Hz / 100 mT, supporting the de-stemming hypothesis on an independent paper from P1. P21's calcium-wave-pattern paper reports stochastic-resonance-style modulation of intracellular Ca^{2+} flux without genomic data. P16 reports cell-line-specific modulation of apoptosis pathway proteins (Bax, Bcl-2 ratios).

K3.6 Mechanism of action emerging within K3

K3 evidence is too thin to articulate a single mechanism, but two distinctive sub-hypotheses emerge: (i) a 1 Hz / 100 mT 'quasi-DC magnetic challenge' regime in which the field is essentially a slowly-modulated SMF and the mechanism resembles macro H plus a slow oscillation — convergent across P6, P13, P25 with 0 healthy harm in 3/3 reported healthy conditions; (ii) a 'patterned biological-information' regime (P21) in which the waveform mimics intercellular calcium-wave-firing and produces selective cancer-cell perturbation at near-zero amplitude (1 μT). The latter is biophysically speculative but the data are non-zero and warrant follow-up.

K3.7 Limitations of K3 evidence

Healthy-cell coverage is small (6 conditions). The under-specified waveforms make replication difficult; the reproducibility of P21's calcium-wave-pattern conditioning depends on the precise temporal pattern, which is not parameterised in the dataset. P15 reports only one row (HT-1197 bladder cancer at 30 Hz / 1.5–16 mT) without a healthy control. The aggregate K3 selectivity index of ≈ 4.2 should therefore be treated as preliminary; the convergent 1 Hz / 100 mT signal across P6, P13, P25 is the most defensible K3 finding.

4. Step 3 — Cross-subgroup synthesis and experimental hypothesis

4.1 Shared mechanistic signatures across macro-categories

Three mechanistic signatures recur across multiple macro-categories and are the strongest candidates for a 'universal' non-thermal antitumor effect.

Signature 1 — Mitochondrial-membrane-potential collapse and ROS rise. $\Delta\Psi_m$ Direction = -1 dominates in macro A (21/21 conditions in nsPEF), is the dominant trend in macro K (8/18 cancer measurements, with the rest split between 0 and +1), is reported in macro G (2/10), macro H (1/1), and macro D (under 'Mentioned'). The corresponding ROS signature (Direction = +1 in cancer) is well established in macro K (10/16) and partially in A, C, F. The combined chain $\Delta\Psi_m\downarrow + ROS\uparrow$ is therefore the single most cross-cutting mechanism in the dataset.

Signature 2 — Cell-cycle G0/G1 arrest. Phase-resolved cell-cycle data are scarce, but where present (macros K with 11 conditions, F with 2 conditions, H with 1 condition, J with 2 conditions, E with 4 conditions), the dominant finding is G0/G1 fraction increase / S fraction decrease, consistent with a mitochondrial-stress $\rightarrow$ p21 $\rightarrow$ cyclin D1 pathway. This convergence across ELF-EMF (K), AMF kHz (F), rotating ELF (J), LTE (E) and SMF (H) is striking given how different the underlying physics are.

Signature 3 — Cytoskeletal / morphological remodelling. Morphology and cytoskeleton flags are positive in the majority of conditions across A, B, C, D, F, J and K. The molecular signatures differ (Rho GTPases in J; tubulin in C; FAK / paxillin in J), suggesting that the upstream trigger is field-specific but the downstream cytoskeletal disruption is convergent. In therapeutic terms, this is encouraging because cytoskeletal disruption is associated with both anti-mitotic and anti-migratory phenotypes.

Figure 6. Direction distribution of cancer-cell outcomes by sub-module
(green = therapeutic direction; grey = no change; red = counter-therapeutic)

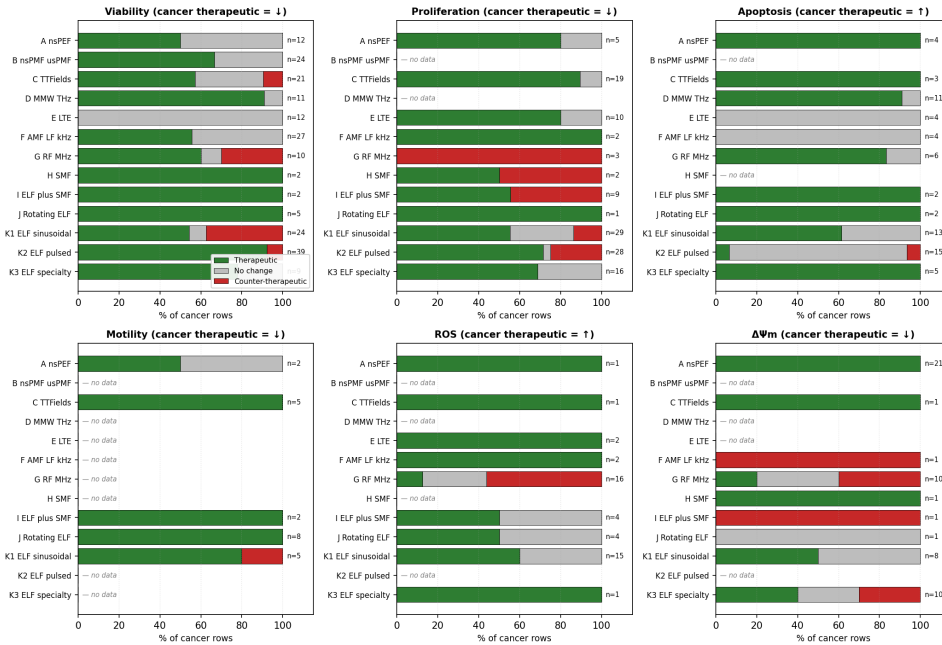
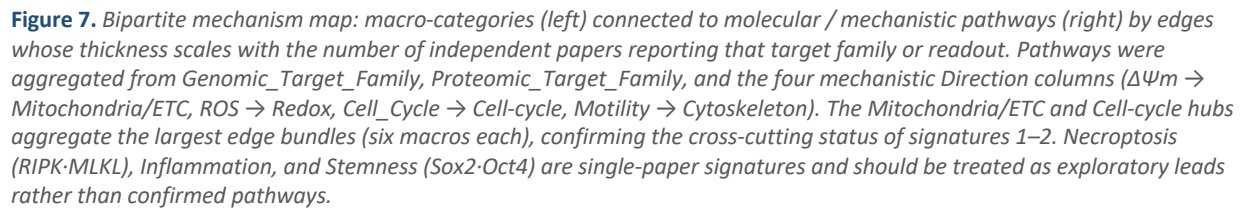


Figure 6. Direction distribution of cancer-cell outcomes by sub-module. Each row is a sub-module; bar segments show the fraction of cancer-cell rows reporting therapeutic, no-change, or counter-therapeutic Direction (the therapeutic direction varies by readout: ↓ for viability/proliferation/motility/ΔΨm, ↑ for apoptosis/ROS). The matrix exposes patterns invisible in any single sub-module: ΔΨm collapse is therapeutically convergent across A, B, C, H (100 %) and partial in K1/K3; macro G (RF MHz) shows the dataset's largest counter-therapeutic ROS fraction (consistent with bidirectional cyclotron-resonance behaviour); macro E (LTE) has zero therapeutic viability effect; K2 ELF pulsed has a near-perfect therapeutic apoptosis fraction (n=15); macro F has 100 % viability therapeutic but with only n=27 measurements. The empty cells denote macros where that outcome is essentially never measured (e.g. motility outside C, J, K1; ΔΨm outside A, G, K1).



Several mechanisms appear in only one macro-category and define the unique therapeutic identity of that macro:

Calcium-cyclotron-resonance modulation (macro G, RF MHz) is a unique mechanism that, at the right (AC, DC) combination, produces apoptosis induction without ROS rise. The bidirectional (cytotoxic vs proliferative) frequency dependence is unique within the dataset and is biophysically rich, but the absence of healthy-cell controls makes selectivity claims premature.

Massive apoptotic and senescent induction at sub-thermal MMW (macro D, Paper 12) is the largest absolute apoptosis signal in the dataset (up to 1542 % of control) and is matched by a complete absence of effect on MCF-10A — the cleanest selectivity signal anywhere.

Resonance-tuned ferroptosis (macro I, Paper 9) is the only place in the dataset where ferroptosis is reported as positive (2 conditions), and it is mechanistically distinct from the mitochondrial-stress chain that dominates elsewhere.

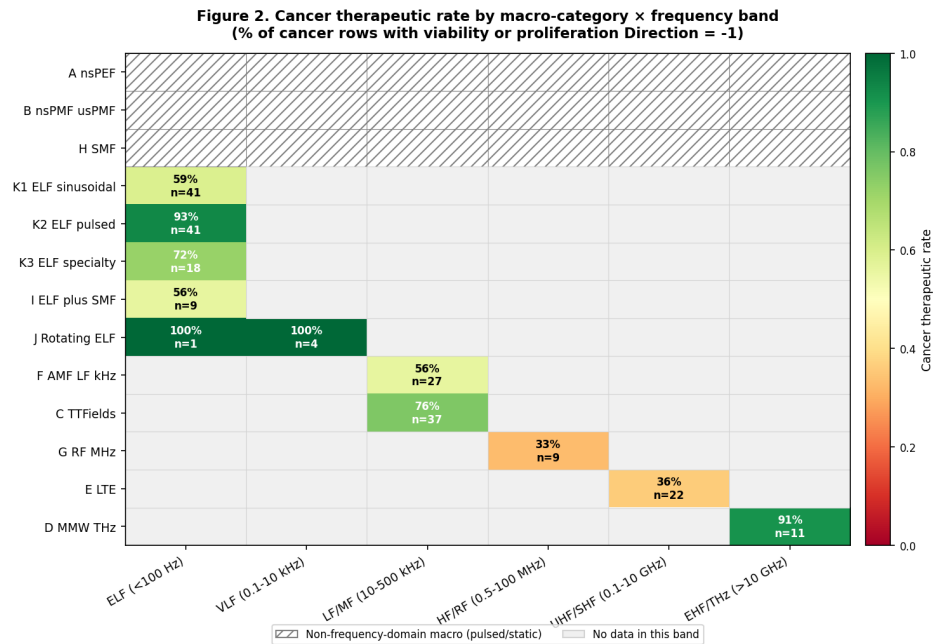


Figure 2. Cancer therapeutic rate (% of cancer-cell rows reporting Direction = -1 for viability or proliferation) by macro-category × frequency band. Macros A, B and H lack a meaningful frequency axis (pulsed-temporal or static fields, hashed) and are shown for completeness. Each macro-category occupies a distinct frequency niche with its own characteristic therapeutic rate, ranging from 33 % (G — RF MHz) to 100 % (J — rotating ELF), supporting the field-physics-driven categorisation rather than a single 'EMF' bin. The 91 % rate at >10 GHz (macro D) and the 93 % rate at <100 Hz pulsed (K2) anchor the high-end of the spectrum, while the 36 % rate at 0.1–10 GHz (macro E, LTE) marks the therapeutically inert band.

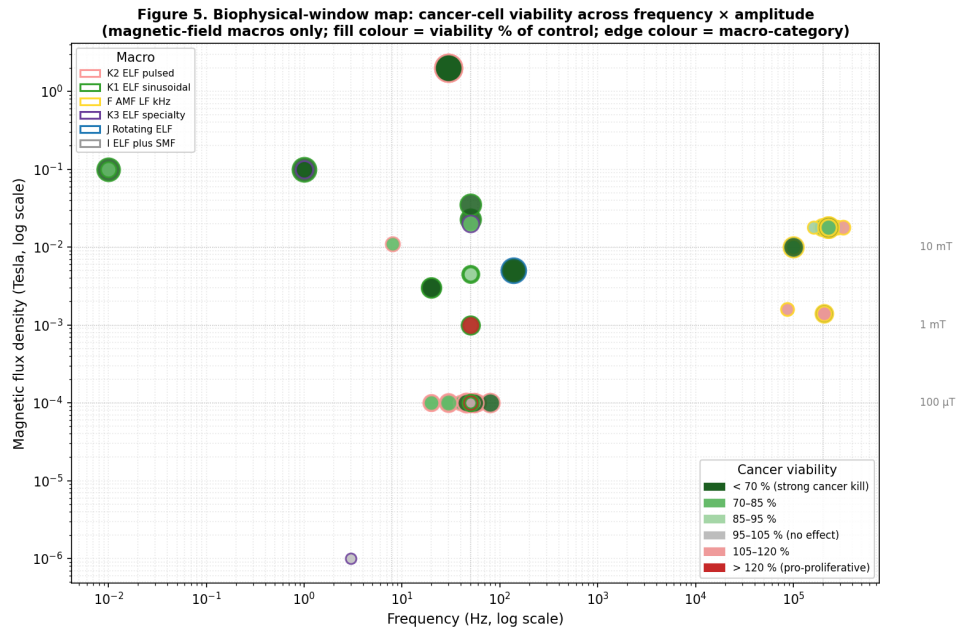


Figure 5. Biophysical-window map for magnetic-field macros (F, I, J, K1, K2, K3): each marker is one cancer-cell row ($n = 103$) plotted on a log-log axis of frequency (Hz) $\times$ magnetic flux density (Tesla). Marker fill colour encodes the cancer-cell viability % of control (dark green ≤ 70 %; mid green 70–85 %; light green 85–95 %; grey 95–105 %; pink 105–120 %; red > 120 %); marker edge colour encodes the macro-category. Two compact biophysical windows emerge: (i) the AMF-kHz cluster at 100–300 kHz $\times$ 1–18 mT (right-hand side, dominated by F yellow edges and dark-to-mid-green fills), the candidate sweet spot of the experimental hypothesis; and (ii) the ELF cluster at 1–80 Hz $\times$ 100 μ T–100 mT (centre-left, dominated by K2 pink and K1 green edges). The single 50 Hz $\times$ 1 mT red marker is Paper 19's hormetic pro-proliferative point. The single 30 Hz $\times$ ≈ 2 T marker is Paper 5's super-strong pulsed regime.

4.3 Tumour-type response patterns

Glioblastoma / glioma is the most-tested tumour type and responds across multiple macro-categories: TTFields (C), AMF kHz (F), rotating ELF (J), and ELF-general (K). It is plausibly the broadest indication for an EMF monotherapy because (i) the mitotic fraction in glioblastoma is high, supporting TTFields-class mechanisms; (ii) the mitochondrial-respiratory-chain dependence is high, supporting AMF kHz; (iii) the post-mitotic surrounding tissue (mature astrocytes and neurons) is intrinsically sparing-friendly. Triple-negative breast cancer (MDA-MB-231) is also broadly responsive (across A, C, F, G, H, I, J, K). Hepatocellular carcinoma (Huh7, Hep3B) is responsive within macro E (LTE) but the response is shared by healthy IMR-90 fibroblasts. Lung cancer (H1299) shows the strongest single response in the dataset (macro D, MMW). Melanoma (A375, B16F10) shows a clean immunogenic-cell-death signature in macros A and C, a finding that is mechanistically attractive but rests on few conditions.

4.4 A testable experimental hypothesis for a future project

Synthesising the dataset, three convergent observations point toward a specific experimental hypothesis. (i) The frequency band 100–300 kHz delivers cancer-cell viability or proliferation reduction in two independent macro-categories (C electric, F magnetic) with selectivity windows demonstrated in independent papers. (ii) Within this band, the magnetic-field route (F, AMF kHz) is more amenable to non-

contact delivery and to depth penetration than the electric route (C, TTFields), and Paper 4's 9-cancer / 3-healthy panel at 227 kHz / 18 mT is the strongest selectivity demonstration for a magnetic-only field. (iii) The mechanistic chain (mitochondrial perturbation → ROS rise → G0/G1 arrest) is consistent across macros C, F, K and would predict that healthy cells with intact mitochondrial reserve and antioxidant buffering would resist the same exposure.

Hypothesis statement. A 200–230 kHz sinusoidal alternating magnetic field at 5–18 mT, applied for 30–60 minutes daily over 5 days, will reduce cancer-cell viability by $\geq 30\%$ and proliferation by $\geq 50\%$ in a panel of glioblastoma (U87, U251, GBM-patient-derived) and triple-negative breast cancer (MDA-MB-231, BT549, HCC38) cell lines, while leaving viability and proliferation of matched healthy lines (NHA, MCF-10A, HUVEC, HCF) unchanged at any of the same parameter combinations.

Primary outcome. Mitochondrial membrane potential ($\Delta\Psi_m$) measured by JC-1 ratio at 0, 6 and 24 h post-exposure, with cell-line-paired ANOVA. Predicted result: $\Delta\Psi_m$ Direction = -1 in cancer lines with effect size $\geq 25\%$, $\Delta\Psi_m$ Direction = 0 in healthy lines.

Secondary outcomes. (a) γ -H2AX foci by immunofluorescence (DNA damage marker; based on macro F genomic targets); (b) ROS measurement by CellROX or DCFH-DA flow cytometry; (c) Cell-cycle phase distribution by PI/FxCycle flow cytometry, expecting G0/G1 increase in cancer lines; (d) Antioxidant pool (GSH/GSSG ratio) — predicted to decrease in cancer cells and remain stable in healthy cells (based on macro F antioxidant block); (e) Caspase-3 cleavage and Annexin-V/PI dual staining at 24 and 48 h to determine death route (apoptotic vs non-apoptotic, the latter expected based on macro F apoptosis Direction = 0).

Selectivity window. Operationally, the selectivity index (SI) is defined as the ratio of cancer-cell viability reduction to healthy-cell viability reduction at the same exposure parameters. We hypothesise $SI \geq 4$ across all 4 healthy-line $\times$ 6 cancer-line pairs at the optimal point (227 kHz, 18 mT, 30 min, 24 h observation).

Mechanistic rationale. The hypothesis rests on the assumption that cancer cells have a distinct mitochondrial vulnerability (high $\Delta\Psi_m$ at baseline, dysregulated respiratory chain, depleted antioxidant reserve in the Warburg phenotype) that is preferentially loaded by induced eddy currents in the 100–300 kHz band. The induced electric-field amplitude E_{ind} inside a cell of radius r exposed to a uniform sinusoidal magnetic field B at frequency f is approximately $E_{ind} \approx \pi \times f \times B \times r$, which at 227 kHz and 18 mT and $r \approx 8 \mu\text{m}$ gives $E_{ind} \approx 0.1 \text{ V/m}$ at the cell membrane — sub-electroporation but sufficient to perturb the inner-mitochondrial-membrane potential of cells whose $\Delta\Psi_m$ is already maximised. Healthy cells, with lower baseline $\Delta\Psi_m$ and intact antioxidant capacity, buffer the perturbation.

Risk and mitigation. The principal risk is loss of selectivity at 18 mT in cell types with high mitotic fraction (e.g. proliferating fibroblasts in vivo). Mitigation is to test additional healthy proliferating controls (HDFa, BJ-5ta) and to reduce amplitude to the 5–10 mT range and increase exposure time to 60 min. A secondary

risk is hormetic pro-proliferative effect at very short exposure or very low amplitude (as observed in macro K Paper 19); mitigation is to include 1 mT and 5 mT amplitude points to map the full dose-response curve.

5. Step 4 — Global limitations and conclusions

5.1 Global limitations of the dataset as a whole

Five global limitations apply across all macro-categories and constrain any meta-conclusion drawn from this dataset. First, coverage imbalance is severe: macro K alone supplies 34 % of conditions and 37 % of papers, while four macro-categories (B, H, I, J) rest on three or fewer papers each. Aggregate statistics across macros are therefore weighted heavily by the largest, most heterogeneous macro-category. Second, healthy-cell sampling is sparse and skewed: 23 of 49 papers (47 %) include no healthy controls at all, and macro categories B, G, H contain zero healthy-cell conditions. The selectivity question — which is the central question of the project — is structurally limited by the available comparators. Third, mechanistic readouts are unevenly distributed: viability and proliferation are densely sampled, but ferroptosis, senescence, antioxidant response, and motility are systematically under-sampled. The dataset is informative about whether cells die but less informative about how. Fourth, heterogeneity of exposure parameters within macro-categories is so wide (eight orders of magnitude in amplitude in macro K, three in frequency) that within-macro pooled statistics need to be interpreted at the level of individual (frequency, amplitude, duration) windows, not at macro level. Fifth, all studies are in vitro: there is no in vivo selectivity confirmation in this dataset, and the gap between an in-vitro selectivity index and a clinical therapeutic index is significant.

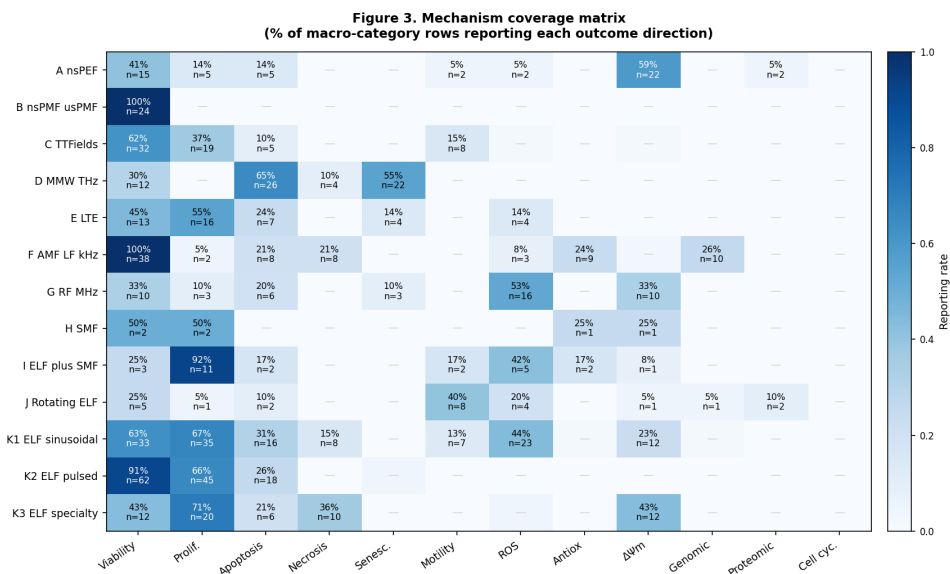


Figure 3. Mechanism coverage matrix: rate at which each sub-module reports a Direction value (i.e. an effect call of -1, 0, or +1) for each of 12 outcomes. Empty cells (—) indicate readouts the macro literature has essentially never measured. The matrix exposes the third global limitation visually: viability and proliferation are densely sampled across nearly all macros, but ROS, antioxidant response, $\Delta\Psi m$, motility, genomic, proteomic, and cell-cycle readouts are sparse, with several macros missing five or more of the twelve readouts entirely. Macro D (MMW/THz) is the only macro reporting senescence at meaningful coverage (55 %); macro G (RF MHz) is the only macro reporting ROS at >50 % coverage; macros A and F dominate $\Delta\Psi m$ and antioxidant respectively. The matrix is the cleanest visual case for the recommendation in section 5.3 that future studies systematically expand mechanistic-readout coverage rather than add yet another viability-and-proliferation paper to a single macro.

5.2 Conclusions

The 49 papers and 434 non-thermal experimental conditions analysed here support a graded conclusion about the therapeutic potential of non-thermal electromagnetic fields as a cancer monotherapy. Figure 1 ranks the 13 sub-modules by selectivity index, colour-coded by the adequacy of healthy-cell controls, and provides the visual anchor for the five graded conclusions below.

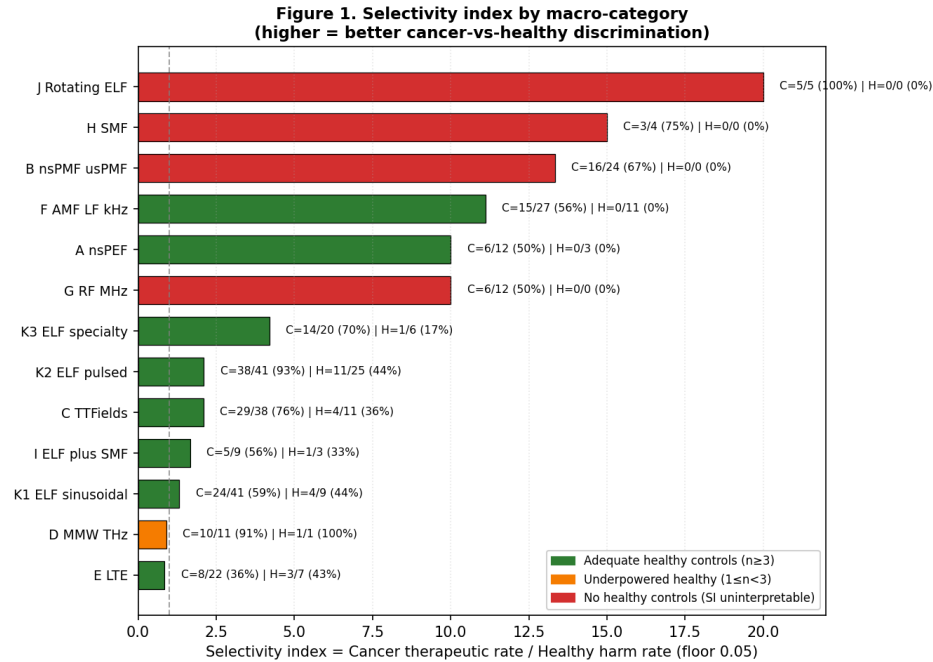


Figure 1. Selectivity index per sub-module, defined as (cancer therapeutic rate) / max(healthy harm rate, 0.05). Cancer therapeutic rate is the fraction of cancer-cell rows with Direction = -1 for viability or proliferation; healthy harm rate is the analogous fraction for healthy cells. Bars to the right of the dashed line ($SI > 1$) indicate cancer-favoured selectivity. Green bars have adequate healthy-cell sampling ($n \geq 3$); orange bars have under-powered healthy sampling ($1 \leq n < 3$); red bars have zero healthy controls and the SI is therefore uninterpretable, flagging an evidence gap rather than a positive selectivity claim. The leading interpretable selectivity signals are F (AMF kHz, $SI = 11.1$, $n = 11$ healthy), K3 (specialty ELF, $SI = 4.2$, $n = 6$ healthy), K2 (pulsed ELF, $SI = 2.1$, $n = 25$ healthy) and C (TTFields, $SI = 2.1$, $n = 11$ healthy).

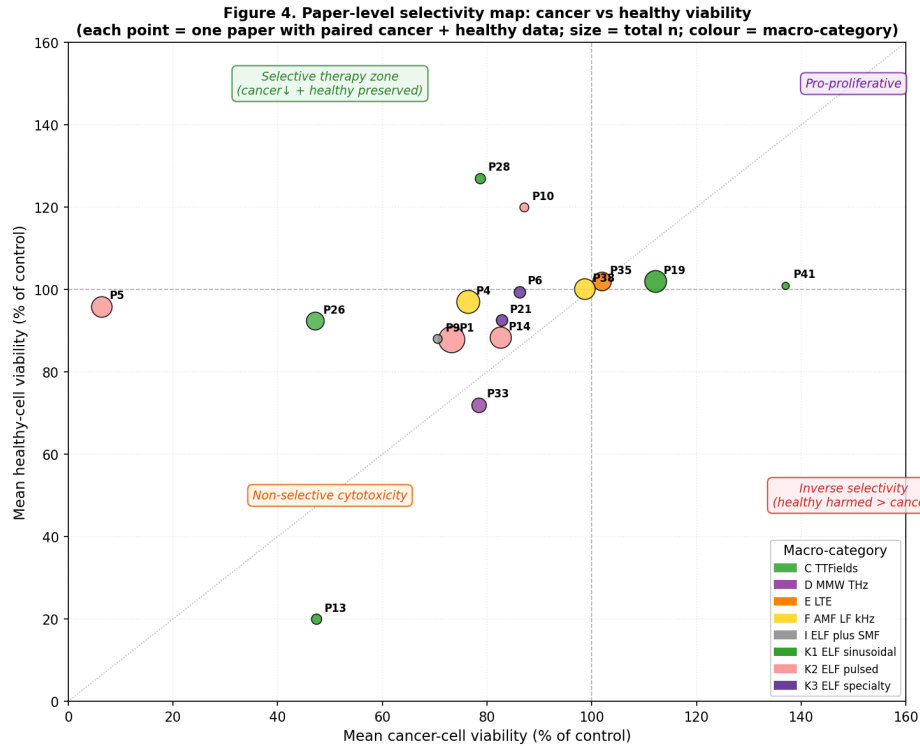


Figure 4. Paper-level selectivity map: each point is one of the 16 papers in the dataset that reports both cancer and healthy viability % of control. X = mean cancer-cell viability; Y = mean healthy-cell viability. Quadrants partition the plane into selective therapy (top-left), pro-proliferative (top-right), inverse selectivity (bottom-right), and non-selective cytotoxicity (bottom-left); the dotted diagonal is the no-selectivity isocline. The strongest selectivity exemplars are P5 (K2 ELF pulsed super-strong, cancer $\approx 6\%$, healthy $\approx 96\%$), P26 (TTFields, cancer $\approx 47\%$, healthy $\approx 92\%$), P4 (AMF kHz, cancer $\approx 76\%$, healthy $\approx 97\%$, the experimental-hypothesis anchor), and P9 (ELF + SMF, cancer $\approx 70\%$, healthy $\approx 88\%$). P19 (K1, cancer $\approx 112\%$, healthy $\approx 102\%$) confirms the pro-proliferative finding at 50 Hz / 0.1 mT. P13 (K1, cancer $\approx 47\%$, healthy $\approx 20\%$) is the dataset's clearest inverse-selectivity outlier and a counter-example to be avoided in protocol design.

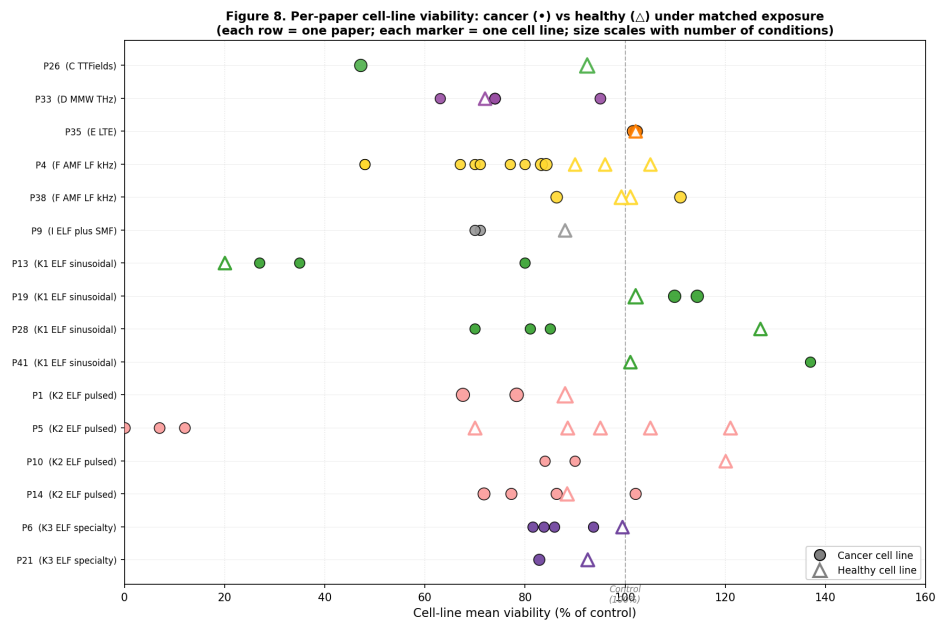


Figure 8. Per-paper cell-line viability: granular view of Figure 4. Each row is a paper; filled circles are cancer cell lines, open triangles are healthy cell lines, with marker size scaling to the number of conditions. Triangles to the right of circles in the same

row indicate selectivity (cancer dropped, healthy preserved); triangles to the left indicate inverse selectivity. The cleanest selectivity panels are P4 (F AMF kHz, 9 cancer circles at 50–88 % vs 3 healthy triangles at 90–105 %), P5 (K2 super-strong, 3 cancer circles at 0–10 % vs 6 healthy triangles spread 70–110 %), and P26 (C TTFields, 1 cancer circle at 47 % vs 1 healthy triangle at 92 %). The K1 ELF sinusoidal cluster (P13, P19, P28, P41) confirms macro K1's heterogeneity: P13 is inverse-selective, P19 is pro-proliferative, P28 has the cleanest selectivity (cancer 79 %, healthy 127 %, $n = 4$), P41 is dose-finding.

First conclusion — clinically credible. TTFields-class intermediate-frequency alternating electric fields (macro C, 100–300 kHz, 1–4 V/cm) have the most robust antitumor evidence in the dataset (proliferation Direction = -1 in 89 % of cancer measurements) and the most robust selectivity evidence (MCF-12 Direction = 0 below 4 V/cm at 150 kHz). This is consistent with the clinical history of TTFields (FDA approval in glioblastoma and pleural mesothelioma) and the dataset adds in-vitro replication across glioblastoma, breast and cholangiocarcinoma. The principal therapeutic question for TTFields-class fields is no longer 'do they work?' but 'where do they fit in the treatment pipeline?'.

Second conclusion — mechanistically promising and underexploited. AMF intermediate-frequency magnetic fields (macro F, 200–230 kHz at 10–18 mT) and W-band MMW radiation (macro D, 75–105 GHz at 0.2 mW/cm²) emerge as the two most attractive, less-clinically-explored candidates for a non-thermal monotherapy. Both deliver clean selectivity in independent papers (Paper 4 across 9 cancer lines vs 3 healthy; Paper 12 across 7 H1299 conditions vs 15 MCF-10A) and both have plausible biophysical mechanisms (induced-eddy-current mitochondrial loading for AMF; biological-window resonance for MMW). The principal next step for both is independent replication and in-vivo extension; the experimental hypothesis articulated in section 4.4 targets the AMF case.

Third conclusion — adjuvant-only. ELF-EMF (macro K) is too parametrically heterogeneous and shows insufficient population-level selectivity (37 % of healthy measurements affected) to support a monotherapy claim at present. It remains a credible adjuvant or sensitiser modality, particularly because of the convergent G0/G1 arrest signature. Specific sub-windows within macro K may be selective (e.g. Paper 19 at 50 Hz / 1 mT for breast cancer), but they need to be defined parameter-by-parameter rather than at the macro-aggregate level.

Fourth conclusion — premature. nsPEF (A), nsPMF (B), RF MHz (G), SMF (H), ELF + SMF (I), and rotating ELF (J) are mechanistically interesting but rest on too few independent papers and / or too few healthy-cell conditions to support either a monotherapy or adjuvant claim. They define rich research questions for the next 5–10 years.

Fifth conclusion — not supported. LTE-modulated UHF radiation (macro E) at SAR ≤ 8 W/kg shows no selective antitumor signal in this dataset. It does generate a low-grade cancer-cell oxidative response but without translating into apoptosis or selective viability reduction, and the proliferation effect is shared with healthy fibroblasts. The dataset does not support repurposing telecom-band exposure as a cancer therapy.

Taken together, the dataset endorses a translational triage in which TTFields-class fields proceed in clinical optimisation, AMF kHz and W-band MMW proceed to large independent replication and matched in-vivo studies, and the remaining macro-categories proceed to mechanistic deepening. The experimental

hypothesis articulated in section 4.4 provides one specific, falsifiable next step in the AMF kHz line of investigation.

5.3 Embedded figures and recommended next-iteration visualisations

Eight figures have been embedded in this report. Figure 1 (section 5.2) ranks the 13 sub-modules by selectivity index with explicit colour-coding of the adequacy of healthy-cell controls. Figure 2 (section 4.2) maps cancer therapeutic rate across macro-categories $\times$ frequency bands, showing the field-physics niche each macro occupies. Figure 3 (section 5.1) is the mechanism coverage matrix, visualising the third global limitation. Figure 4 (section 5.2) is the paper-level cancer-vs-healthy viability scatter that surfaces selectivity exemplars (P4, P5, P26) and counter-examples (P13, P19) inside their macro context. Figure 5 (section 4.2) is the log-log frequency $\times$ amplitude bubble plot for magnetic-field macros, exposing the two compact biophysical windows (AMF kHz at 100–300 kHz / 1–18 mT; ELF at 1–80 Hz / 100 μ T–100 mT). Figure 6 (section 4.1) is the cancer-cell direction-distribution stacked-bar matrix per outcome, exposing patterns invisible at the macro-aggregate level (e.g. macro G's bidirectional ROS, macro E's flat viability, macro K2's near-perfect therapeutic apoptosis). Figure 7 (section 4.1) is the bipartite macro $\rightarrow$ mechanism map, showing Mitochondria/ETC and Cell-cycle as the two convergent hubs. Figure 8 (section 5.2) is the cell-line-resolved selectivity dot plot — the granular companion to Figure 4.

Two further visualisations would help the next iteration but require additional data extraction beyond the 132 columns currently available. (a) A protocol-comparability matrix mapping (frequency, amplitude, duration, waveform) onto a parametric grid to triage which paper-vs-paper comparisons are biophysically valid. (b) A time-course / dose-response micro-plot for each of the dataset's small set of papers that include matched titrations (P4, P19, P26, P14), which would let the field move from binary Direction calls to inflection-point estimates.

— End of report —